2026-2학기 세포의 분자적 이해

Chapter 8
RNA Synthesis and Processing 1

분자의학교실 장병철

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Today's Lecture Contents

1. Introduction about learning performance of today's lecture
2. Overview about gene expression (its definition, process, and function)
3. Transcription in prokaryotes
   - 3.1. *E. coli* RNA polymerase (pol) and its components/role
   - 3.2. Promoter and its role in gene expression
   - 3.3. Transcription by *E. coli* RNA pol (Initiation, Elongation, Termination)
4. Transcriptional regulation in prokaryotes
   - 4.1. Lactose metabolism
   - 4.2. Lactose operon

Learning Performance (학습 성과)

1) 유전자 발현 (gene expression)에서 전사 (transcription) 및 번역 (translation) 과정을 설명할 수 있다.
2) 원핵세포 (procaryotes)에서의 전사를 설명할 수 있다.
3) 원핵세포에서 전사에 관여하는 RNA polymerase (RNA 중합효소) 및 구성성분과 기능을 설명할 수 있다.
4) 원핵세포 전사에서 유전자 promoter의 기능을 설명할 수 있다.
5) 원핵세포 전사 조절기전으로 lactose operon을 설명할 수 있다.

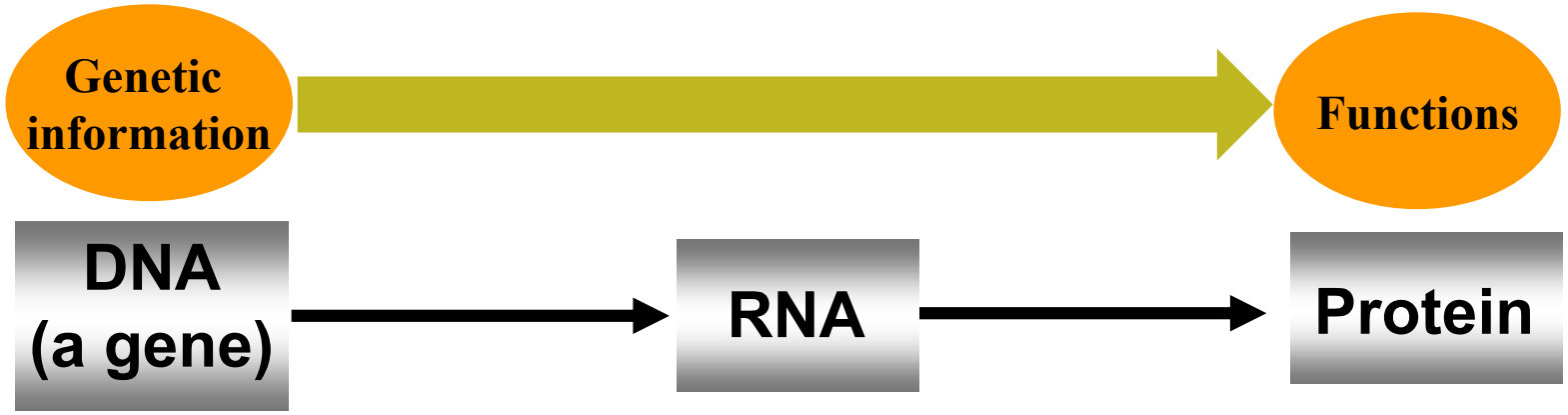
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2. Overview about gene expression (definition, process, and function)

2.1. The definition of gene expression

- The process by which information from a gene is used in the synthesis of a functional gene product (→ "Transfer of genetic information").
- The gene products are often proteins. But, in the case of non-protein coding genes, such as ribosomal RNA (rRNA), transfer RNA (tRNA) or small nuclear RNA (snRNA) genes, the product is a functional RNA (from Wikipedia).

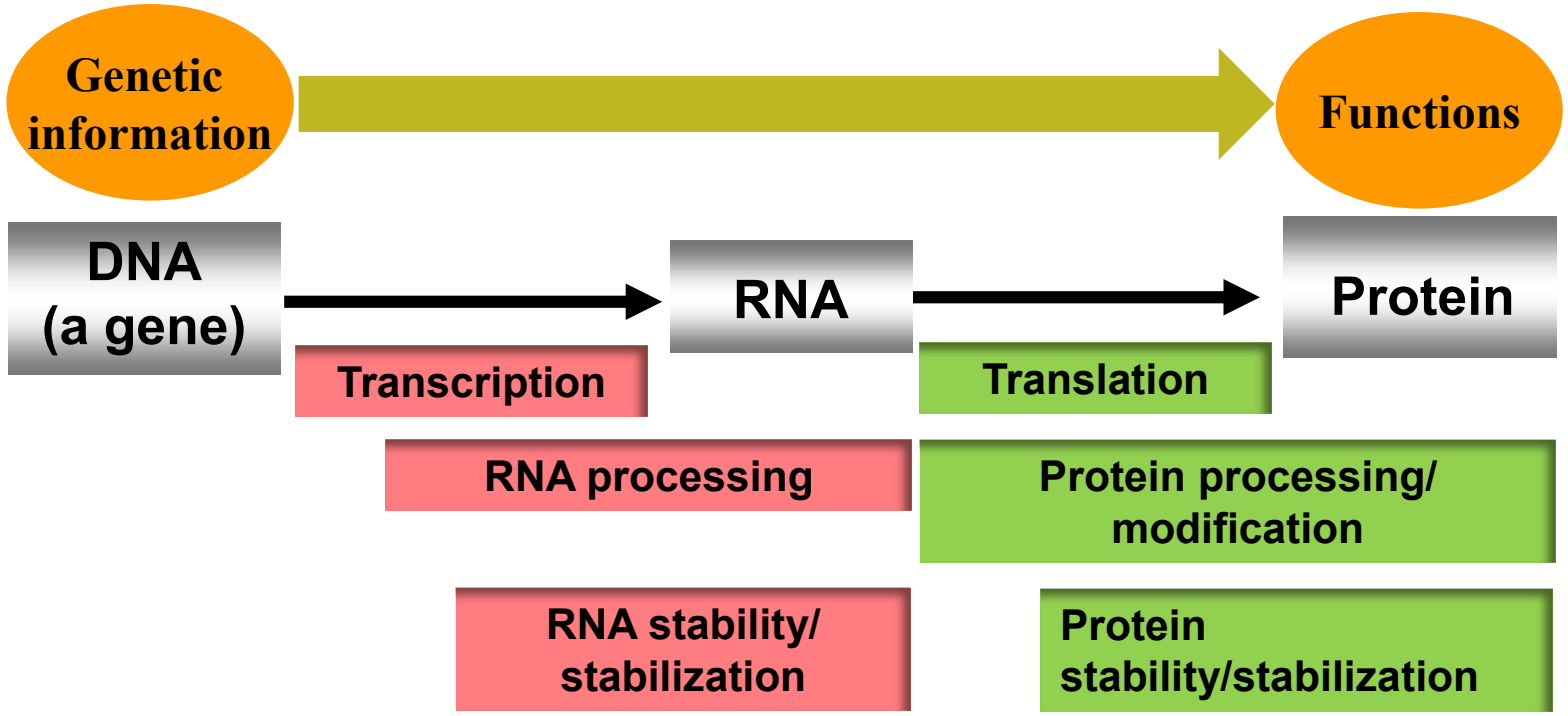


2. Overview about gene expression (definition, process, and function)

2.2. The process of gene expression

√ **Multiple steps in gene expression**

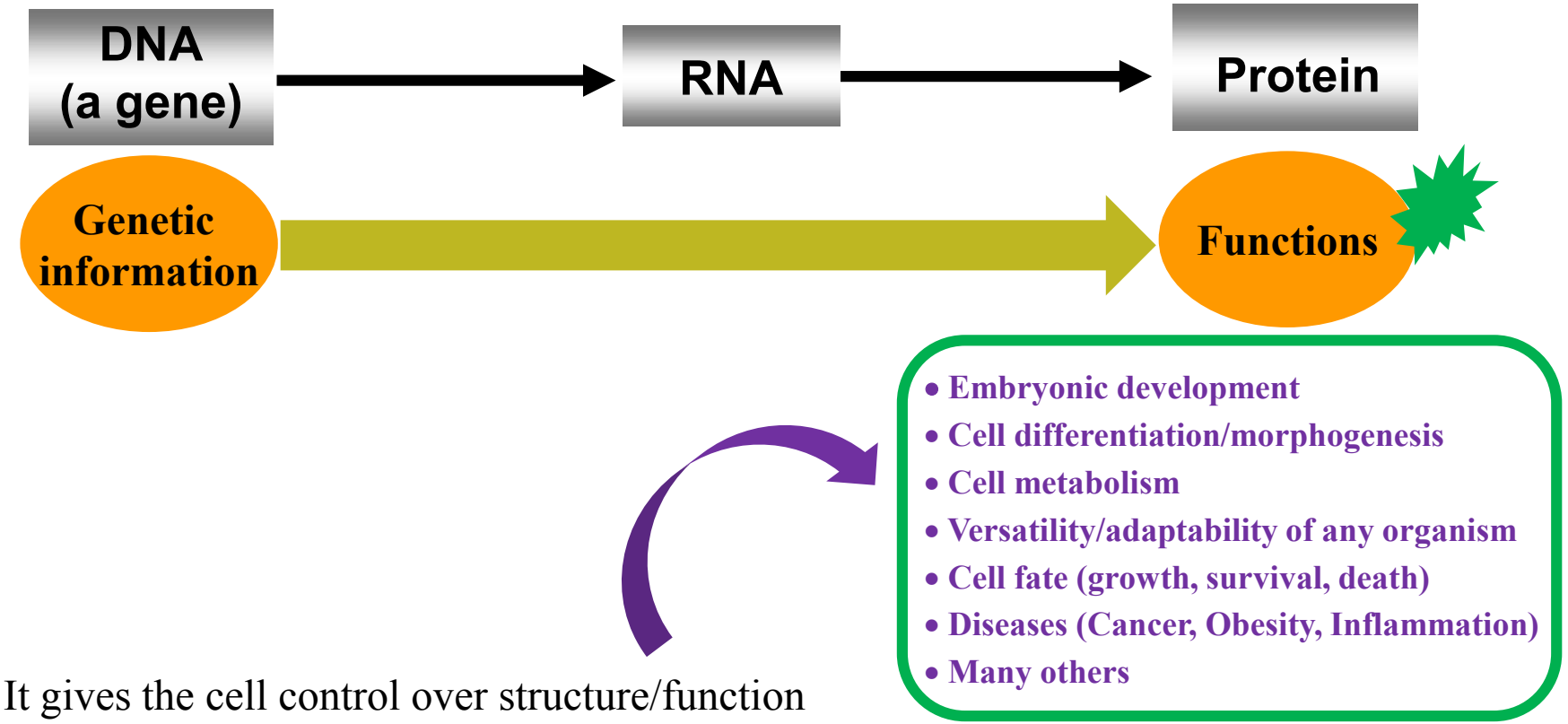
- Transcription, RNA processing, RNA turnover
- Translation, Protein processing/modification, Protein turnover



Many regulatory points in gene expression

2. Overview about gene expression (definition, process, and function)

2.3. The function of gene expression



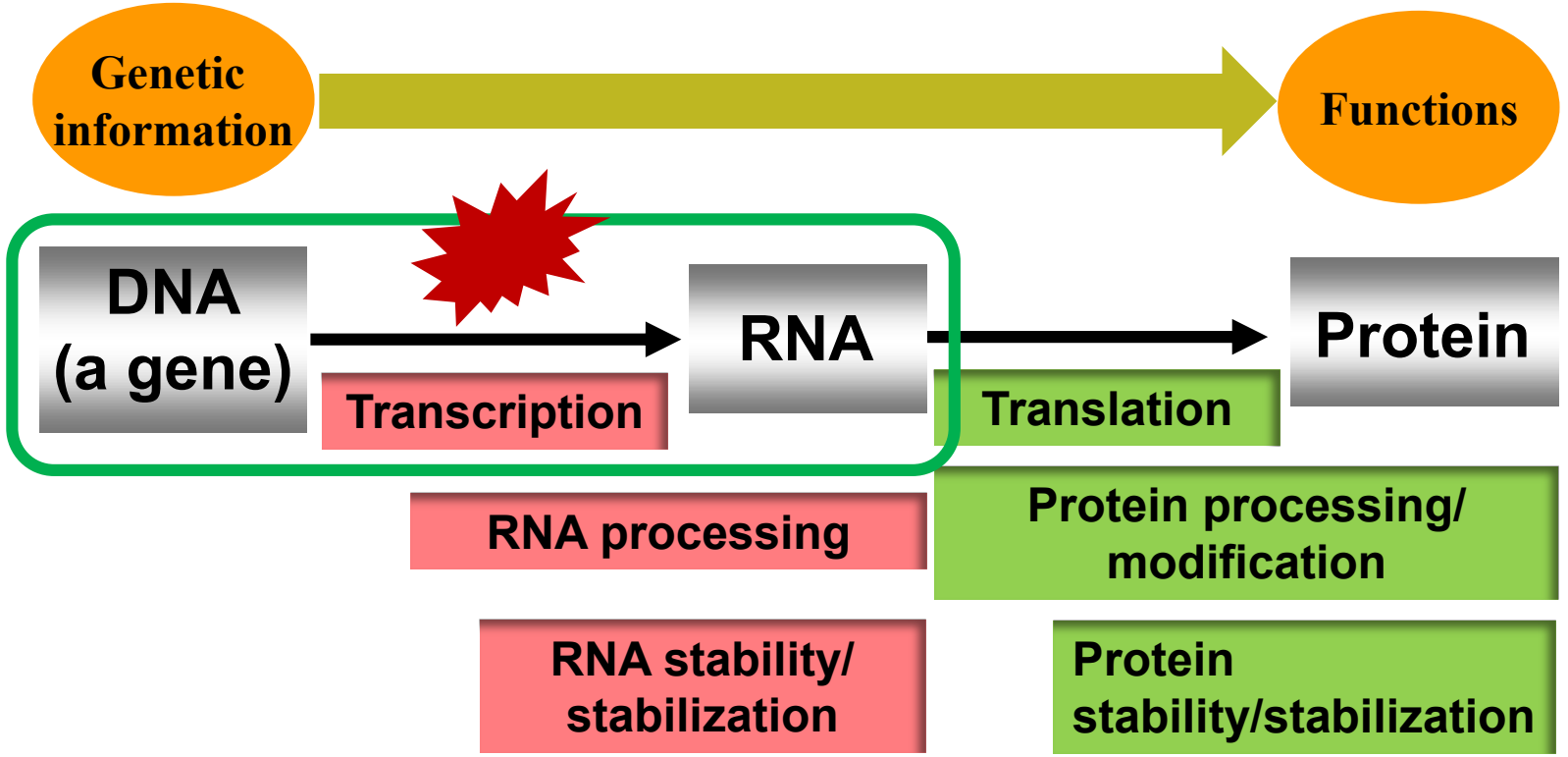
- It gives the cell control over structure/function
- It is the basis for the cellular differentiation, morphogenesis and the versatility and adaptability of any organism
- In genetics, gene expression is the most fundamental level at which the genotype gives rise to the phenotype.
- It allows the cell to adapt to environmental changes and stresses

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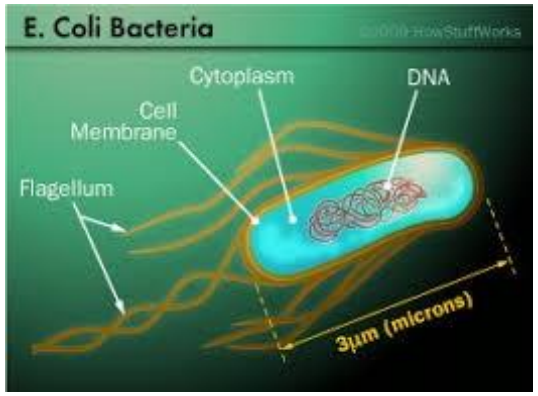
3. Transcription in prokaryotes

- It is the first step in gene expression in which a gene (DNA) is transcribed into RNA by the action of RNA polymerase.
- It is the initial level at which gene expression is regulated.



3. Transcription in prokaryotes

3.1. *E. coli* RNA polymerase (pol): its components and role



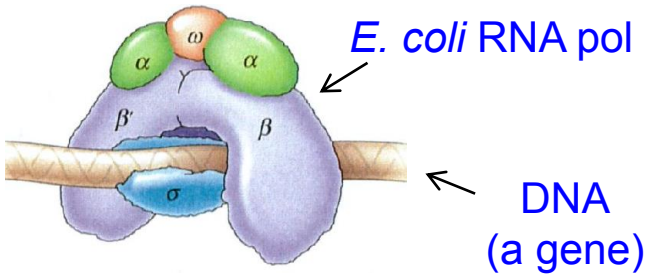
- ***E. coli***

- A prokaryote, a gram-negative bacteria
- Has provided the model for subsequent investigations of transcription in eukaryotic cells

- ***E. coli* RNA polymerase (pol)**

- The principle enzyme responsible for RNA synthesis in *E. coli*

- Pol is a complex enzyme made up of 5 different types of polypeptide chains (subunits) (called α, β, β', ω and σ).



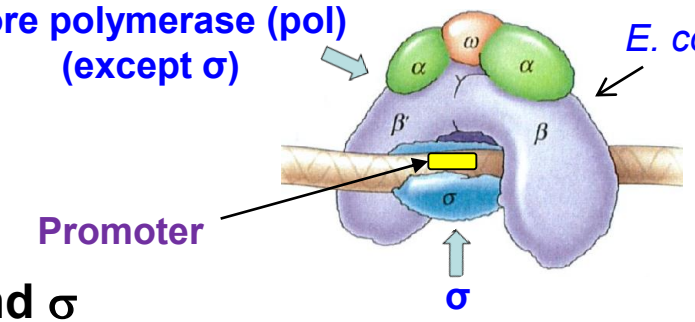
<Figure 8.1. *E. coli* Pol >

- Catalyzes the polymerization of ribonucleoside 5′-triphosphates (NTPs), the growth of RNA chains, as directed by a DNA template, always in the 5′ to 3′ direction.

3. Transcription in prokaryotes

3.1. *E. coli* RNA polymerase (pol): its components and role (Cont'd)

- **The components of *E. coli* RNA polymerase**
  - **Core polymerase (pol)**
  - **σ (sigma)**

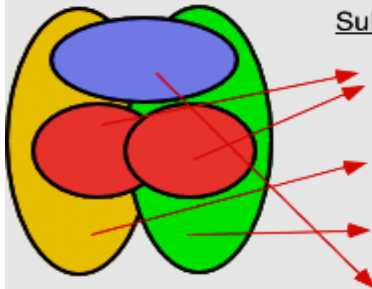


< Figure 8.1. *E. coli* Pol >

- **The role of core pol (pol) and σ**

- **Core pol** is fully capable of catalyzing the polymerization of NTPs into RNA. It does not bind to the DNA sequence (eg, promoter) that signals the initiation of transcription.
- **σ** (sigma) is required to identify the DNA sequence (promoter) for transcription initiation.

Prokaryotic RNA Polymerase: Holoenzyme Enzyme



Subunit	Size	#/Molecule	Function
α	36.5 kD	2	chain initiation and interaction with regulatory proteins
β	151 kD	1	chain initiation and elongation, nucleotide (substrate) binding
β'	155 kD	1	DNA binding
σ	70 kD	1	promoter recognition
ω	10 kD	1	role in enzyme assembly

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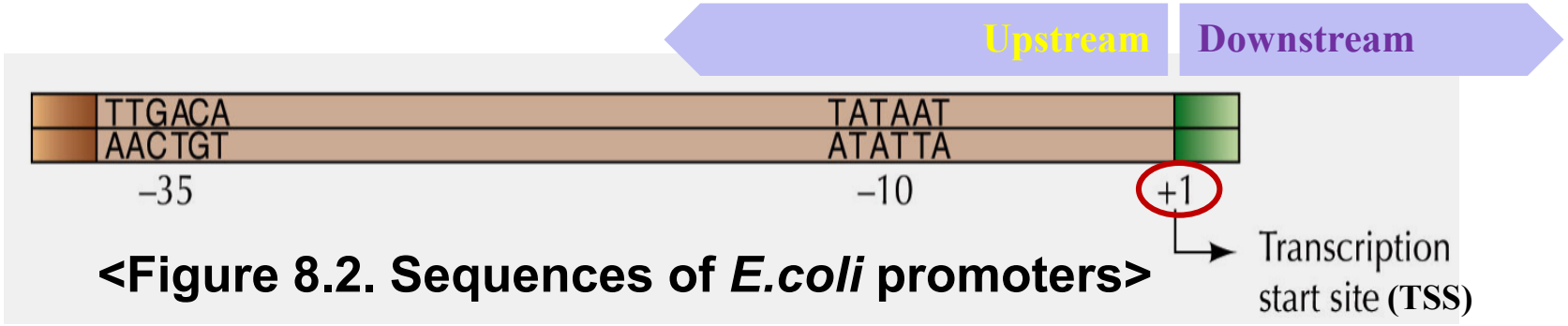
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3. Transcription in prokaryotes

3.2. Promoter

- It is the DNA (a gene) sequence to which RNA pol (sigma subunit) binds to initiate transcription of a gene.

- In general, it has six nucleotides that locate at 10 and 35 base pair (bp) upstream of the transcription start site (TSS).



<Figure 8.2. Sequences of *E.coli* promoters>

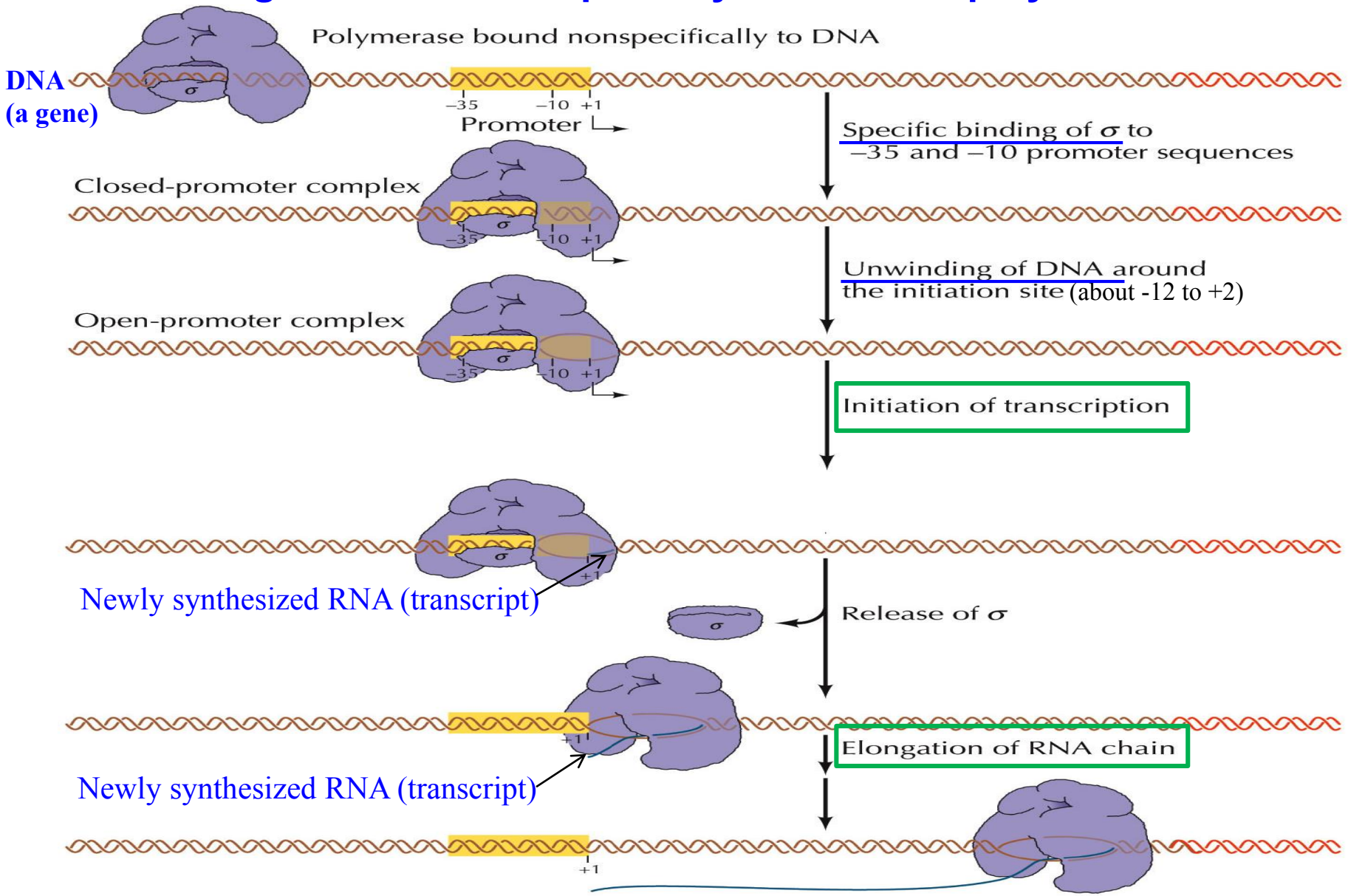
- Consensus sequences are the bases most frequently found at each position in different promoters of genes.

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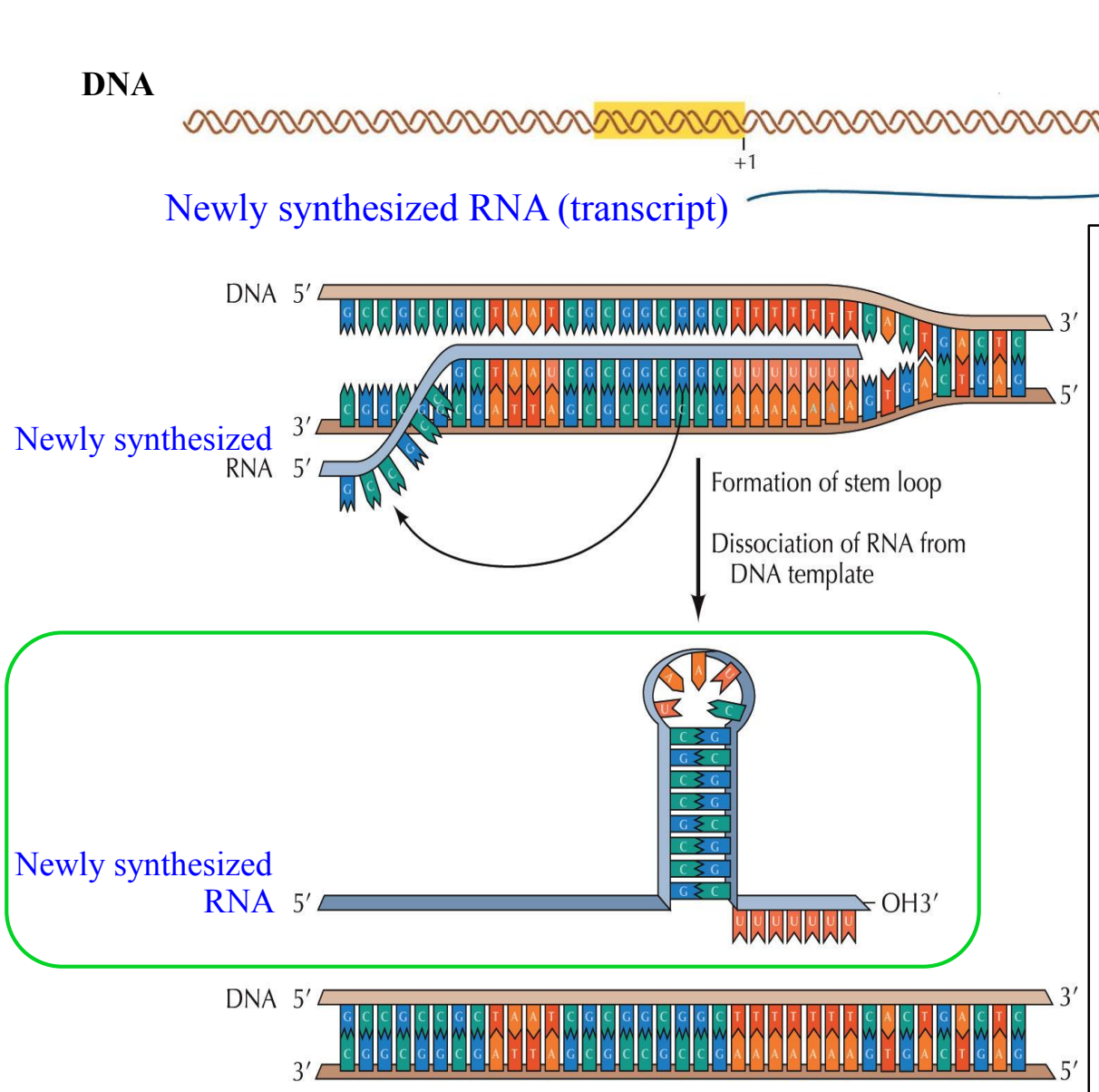
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3. Transcription in Prokaryotes (initiation, elongation, termination)

< Figure 8.3. Transcription by *E. coli* RNA polymerase >



3. Transcription in Prokaryotes (initiation, elongation, termination)



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<Figure 8.5. Transcription termination>

- RNA synthesis continues until the polymerase encounters a termination signal in the DNA template.
- The most common termination signal in the DNA template is a symmetrical inverted repeat of a GC-rich sequence followed by seven A residues
- Transcription of the GC-rich inverted repeat results in a segment of RNA that forms a stable stem-loop structure, which disrupts its association with the DNA template and terminates transcription.

♦ Alternatively, transcription of some genes is also terminated by a specific termination protein (Rho), which binds extended segments (> 60 nts) of single-stranded RNA.

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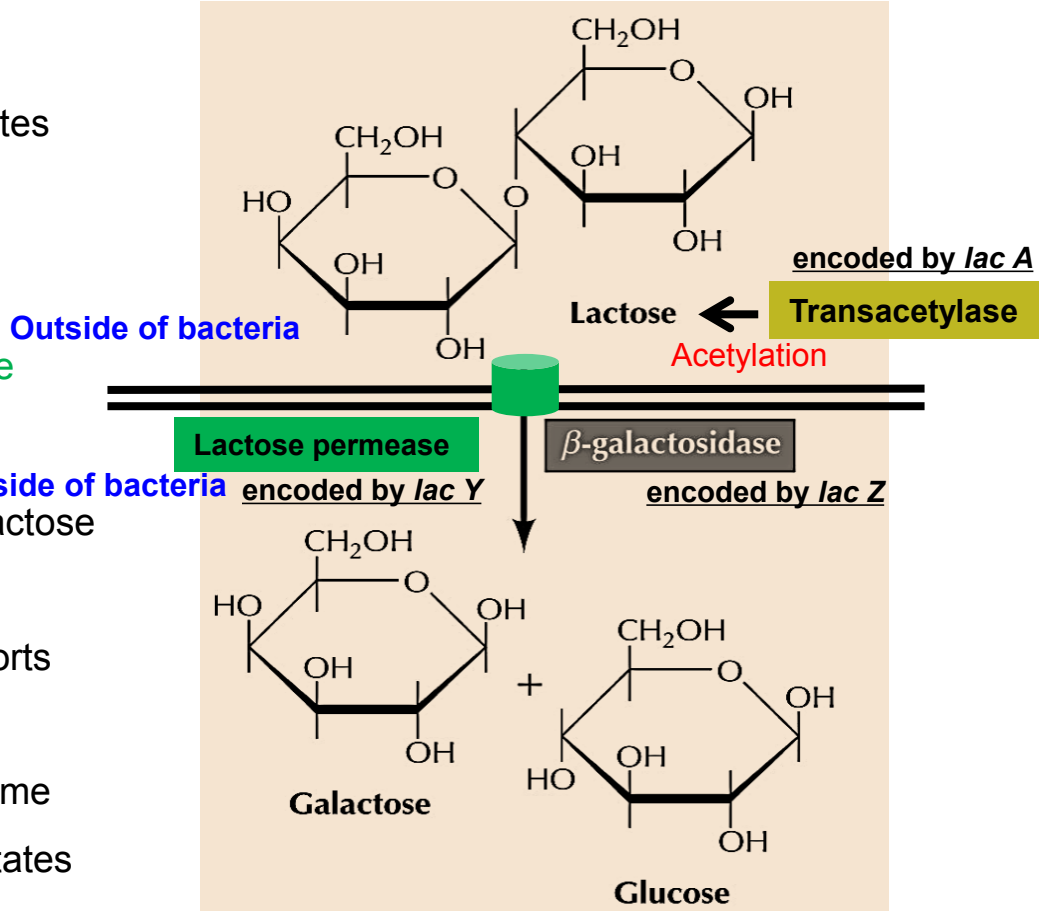
4. Transcriptional regulation in prokaryotes

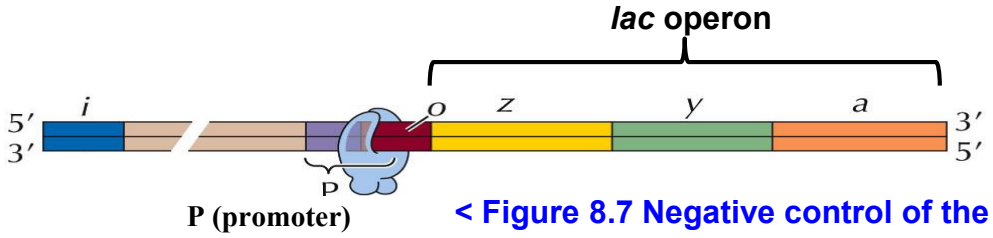
* **Lactose** - used as a source of carbon and energy

4.1. Lactose metabolism

- Most transcriptional regulation in bacteria operates at initiation.
- Studies of gene regulation in the 1950s used the enzymes involved in lactose metabolism (herein lactose catabolism to breakdown lactose to yield energy necessary for their growth).

- ***β*-galactosidase (encoded by *lac Z*)** cleaves lactose into glucose and galactose.
- **Lactose permease (encoded by *lac Y*)** transports lactose into the bacteria.
- **Transacetylase (encoded by *lac A*)** is an enzyme to attach an acetyl group to lactose, which facilitates transport of lactose into the bacteria.





< Figure 8.7 Negative control of the *lac* operon >

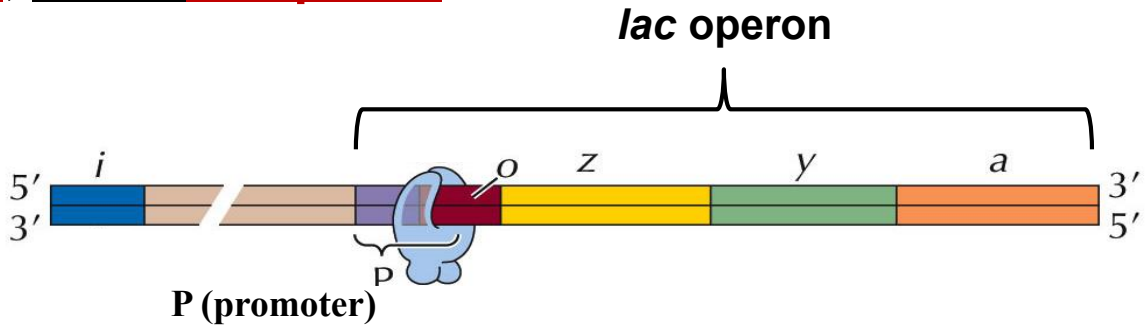
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4. Transcriptional regulation in prokaryotes

4-2. Lactose operon

- **Genes** encoding the enzymes involved in lactose metabolism are expressed as a single unit, called an **operon**.



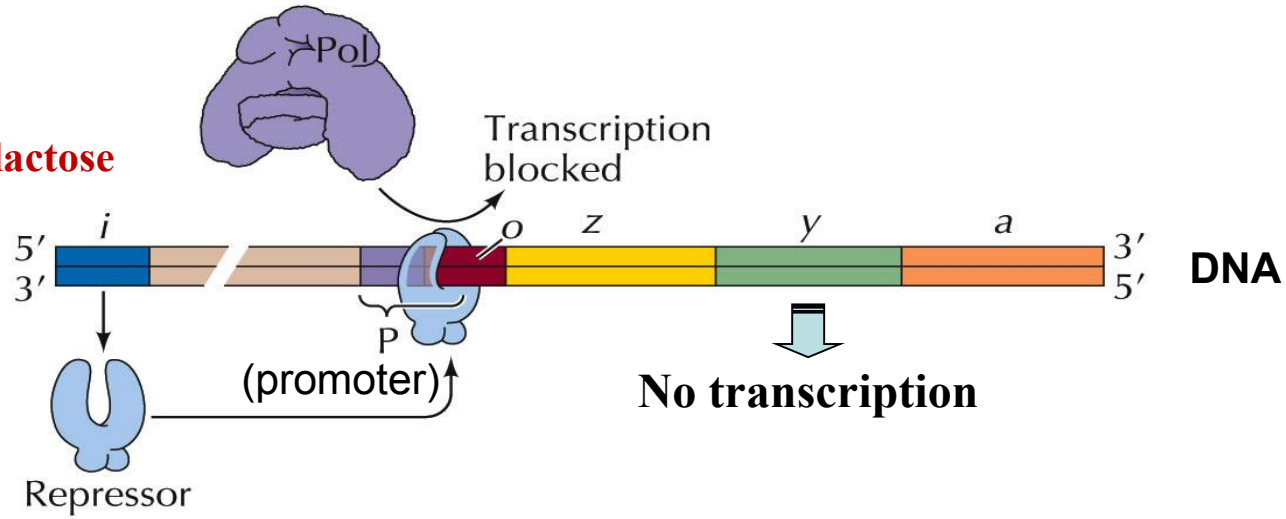
< Figure 8.7 Negative control of the *lac* operon >

- Transcription of the *lac* operon is controlled by ***o*** **(operator, a segment of DNA (20 bp)), adjacent to the transcription initiation site (TIS).**

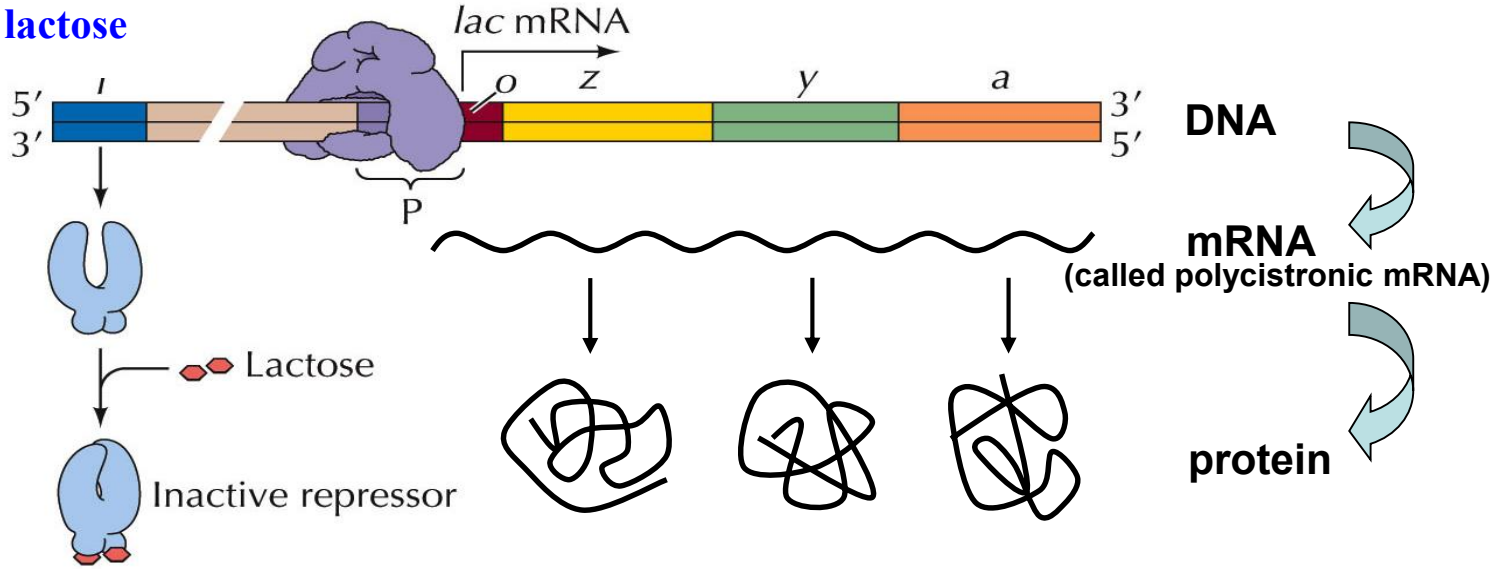
- The ***i*** gene (**not physically linked to the operon**) encodes a protein, called **repressor**, **that specifically binds to the operator *o***.

4. Transcriptional regulation in Prokaryotes

1) In the absence of lactose



2) In the presence of lactose



< Figure 8.7 Negative control of the *lac* operon >

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Lactose can be used as a source of carbon and energy by the action of these enzymes!

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Chapter 8
RNA Synthesis and Processing 2,3

- Transcription and its regulation in eukaryotes
- RNA processing and RNA turnover in eukaryotes

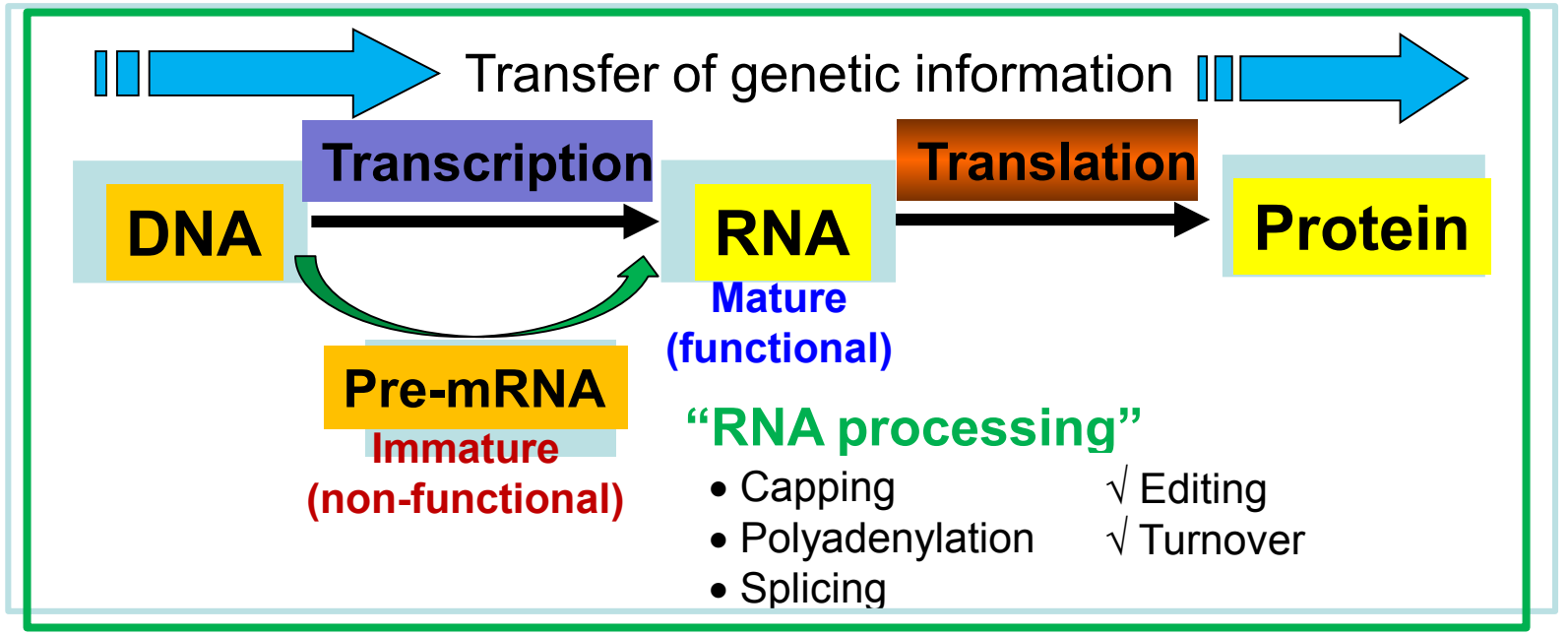
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Chapter 8. RNA Synthesis and Processing 2,3

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1. Introduction about learning performance of today's lecture
2. Transcription and its regulation in eukaryotes
3. Major RNA processing in eukaryotes
4. RNA turnover in eukaryotes



Chapter 8. RNA Synthesis and Processing 2,3

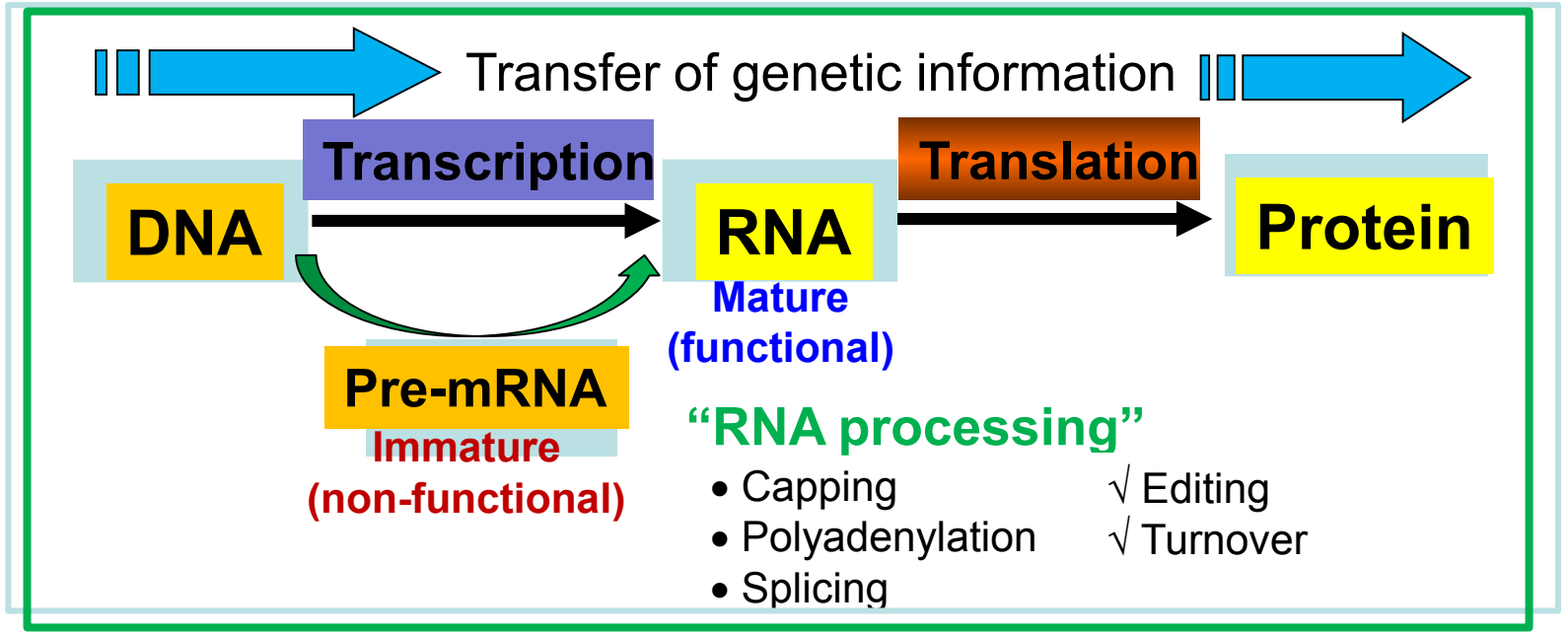
Learning Performance

1. Be able to explain **transcription in eukaryotic cells**.
2. Be able to describe the **types and functions of RNA polymerases** involved in eukaryotic transcription.
3. Be able to explain the **types and functions of transcription factors** in eukaryotic transcription.
4. Be able to describe the **types and functions of cis-acting elements** within genes in eukaryotic transcription.
5. Be able to explain the **mechanisms of transcriptional regulation via chromatin structure and epigenetics** in eukaryotic cells.
6. Be able to explain the **major RNA processing events** in eukaryotic cells.
7. Be able to explain **RNA turnover** in eukaryotic cells.

Chapter 8. RNA Synthesis and Processing 2,3

Lecture Contents

1. Introduction about learning performance of today's lecture
2. **Transcription and its regulation in eukaryotes**
3. Major RNA processing in eukaryotes
4. RNA turnover in eukaryotes



Chapter 8. RNA Synthesis and Processing 2,3

2. Transcription and its regulation in eukaryotes

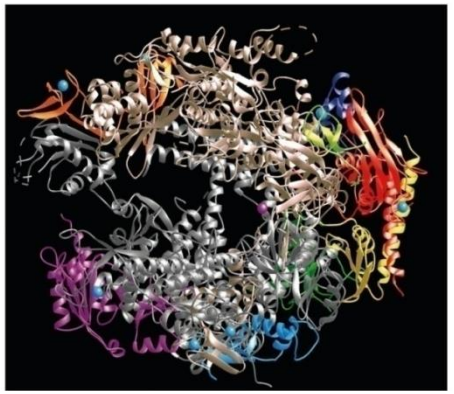
2.1. RNA polymerase(s) in eukaryotes

- Eukaryotic cells have three RNA polymerases (I, II, III) that transcribe different classes of genes in the nucleus.
- They are complex enzymes, consisting of 12 to 17 different subunits each.
- They all have 9 conserved subunits, 5 of which are related to α, β, β', and ω subunits of bacterial RNA pol.

Table 8.1 Classes of Genes Transcribed by Eukaryotic RNA Polymerases

Type of RNA synthesized	RNA polymerase
Nuclear genes	
mRNA	II
miRNA	II
lncRNA	II
tRNA	III
rRNA	
5.8S, 18S, 28S	I
5S	III
snRNA and scRNA	II and III[a]
Mitochondrial genes	Mitochondrial[b]
Chloroplast genes	Chloroplast[b]

[a]Some small nuclear (sn) and small cytoplasmic (sc) RNAs are transcribed by polymerase II and others by polymerase III.
[b]The mitochondrial and chloroplast RNA polymerases are similar to bacterial enzymes.



<Figure 8.9 Structure of yeast RNA pol II>

Chapter 8. RNA Synthesis and Processing 2,3

2. Transcription and its regulation in eukaryotes

2.2. RNA polymerase II and transcription factors (TFs) in eukaryotes

- **RNA polymerase II in eukaryotes**

  - is responsible for synthesis of mRNA.
  - Unlike prokaryotic RNA polymerase, it **requires initiation factors that** (in contrast to bacterial σ factors) **are not associated with the polymerase**. One of them identified in the nucleus is **transcription factor (TF)** that binds **promoter region(s) of a gene**.

- **Types of TFs in eukaryotes**

  (1) General TFs (also known as basal TFs)

  - are proteins that bind to general promoters (general DNA sequences), either alone or with other proteins in a complex, to control the rate of transcription (initiation) of a gene (DNA).
  - constitute part of the basic transcription machinery.

  (2) Specific TFs

  - are proteins involved in stimulation of transcription (initiation) of a gene containing the specific regulatory DNA sequence such as enhancer

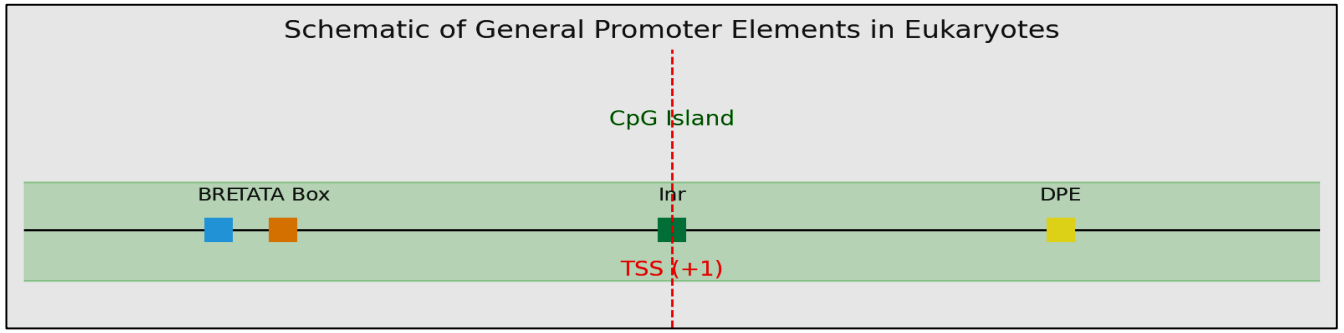
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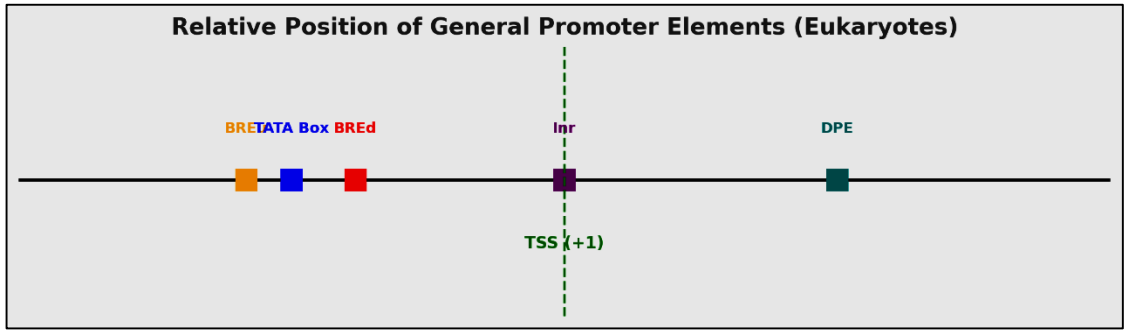
2. Transcription and its regulation in eukaryotes

2.2. RNA polymerase II and transcription factors (TFs) in eukaryotes

► General Promoter (DNA Sequences) and their general TFs in Eukaryotes

General DNA Sequence (Promoter Element)	Consensus Sequence	Binding General Transcription Factor	Function
TATA Box	TATAAA (~ -25 to -30 bp upstream of TSS (+))	**TBP** (TATA-binding protein, part of TFIID)	Positions RNA polymerase II at the correct transcription start site
Initiator (Inr)	YYANWYY (includes TSS)	**TFIID (TAFs)**	Defines transcription start site; works with or without TATA box
BRE (TFIIB Recognition Element)	SSRCGCC (~ -37 to 32 bp, ~-23 to -17 bp of TSS)	**TFIIB**	Provides binding site for TFIIB; stabilizes TBP-DNA interaction and recruits RNA pol II
DPE (Downstream Promoter Element)	RGWYV (~ +28 to +32 downstream of TSS)	**TFIID (TAFs)**	Functions in TATA-less promoters together with Inr; helps recruit RNA pol II
CpG Islands	GC-rich region (~1 kb)	Indirect (via Sp1 and other TFs, then GTFs)	Found in housekeeping genes; epigenetic regulation via DNA methylation





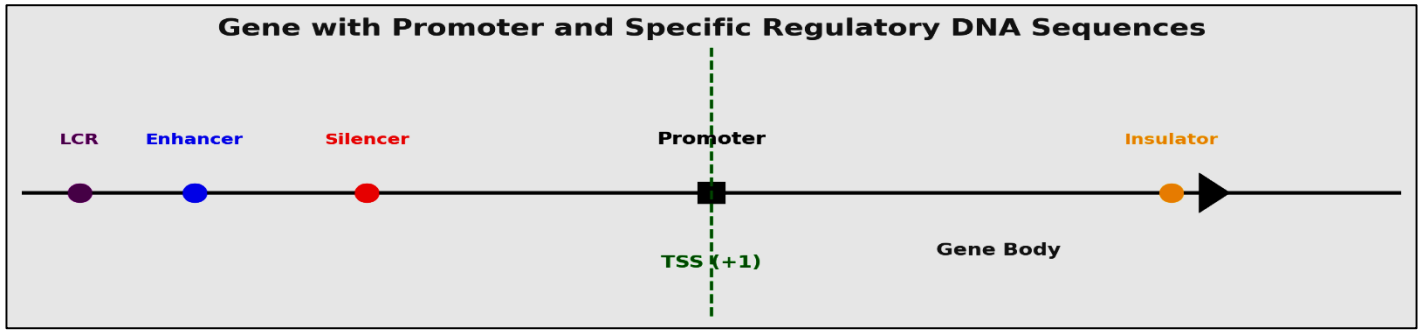
Chapter 8. RNA Synthesis and Processing 2,3

2. Transcription and its regulation in eukaryotes

2.2. RNA polymerase II and transcription factors (TFs) in eukaryotes

► ► Specific Regulatory DNA Sequences (Enhancer) and Their Specific TFs in Eukaryotes

Enhancer	Binding Transcription Factors (TFs)	Function
Immunoglobulin Heavy Chain (IgH) Enhancer	NF-κB, E2A, PU.1	B-cell specific activation; drives high-level antibody gen e expression
β-globin Locus Enhancer (LCR)	GATA-1, NF-E2, KLF1 (EKLF)	Erythroid-specific regulation; controls β-globin expressi on during development
SV40 Viral Enhancer	AP-1 (Fos/Jun complex), NF-κB, Sp1	Strong, non-specific enhancer; widely used in experime ntal gene expression systems
Interferon-β (IFN-β) Enhancer / Enhanceosome	NF-κB, IRF-3/7, ATF-2/c-Jun	Inducible enhancer; activates IFN-β transcription during viral infection
Albumin Enhancer	HNF-1, HNF-3, HNF-4 (Hepatocyte Nuclear Factors)	Liver-specific expression; ensures albumin synthesis in hepatocytes

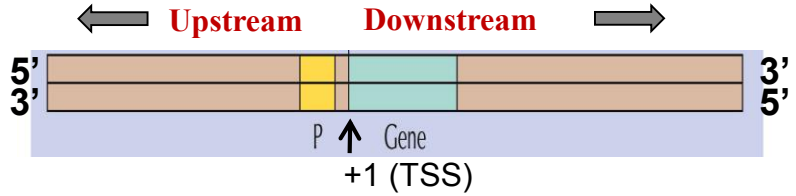


Chapter 8. RNA Synthesis and Processing 2,3

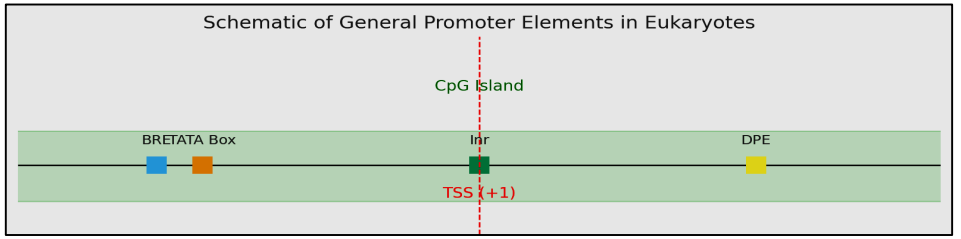
2. Transcription and its regulation in eukaryotes

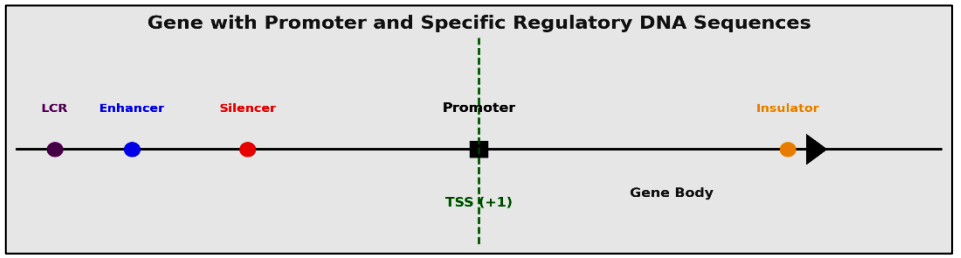
2.2. RNA polymerase II and transcription factors (TFs) in eukaryotes (cont'd)

- **General TFs** (TATA box is one of general promoters)
  - The most common: **TFIIA, TFIIB, TFIID, TFIIE, TFIIF, TFIIH, TABP** (TATA box binding protein, called TBP)
  - **Some of them bind promoter directly, but many of them do not bind; Instead, they become a part of the large transcription initiation complex (TIC) that interacts with RNA pol II** via **protein-protein interaction (PPI).**
  - Are **necessary for a basal transcription of a gene to occur**
  - **bind approx. -1 to -34 promoter region of a gene**



- **Specific TFs**
  - **bind approx. -34 to -250 to further upstream**
  - **has the specificity to bind specific regulatory DNA seq.**
  - **has the activity to stimulate (increase) transcription of a gene containing the specific regulatory DNA seq.**





Chapter 8. RNA Synthesis and Processing 2,3

2. Transcription and its regulation in eukaryotes

2.2. RNA polymerase II and transcription factors (TFs) in eukaryotes (cont'd)

► **Locations of Specific DNA Regulatory Sequences (Enhancer, etc) in Eukaryotes**

Regulatory Element	Typical Location (relative to promoter/TSS)	Notes
Enhancer	Can be **far upstream**, **downstream**, or even **within introns** (up to tens–hundreds of kb away)	Acts via **DNA looping** to interact with promoter and recruit transcription machinery
Silencer	Upstream, downstream, or intragenic (sometimes close to promoter)	Repressors bind → suppress transcription, often by recruiting **HDACs** or altering chromatin
Insulator	Positioned **between enhancer and promoter** or at **chromatin domain boundaries**	Blocks inappropriate enhancer–promoter interaction; organizes chromatin loops
Locus Control Region (LCR)	**Several kb upstream** of gene cluster (e.g., β-globin cluster)	Contains multiple enhancers; ensures **long-range, tissue-specific, high-level expression**

Chapter 8. RNA Synthesis and Processing 2,3

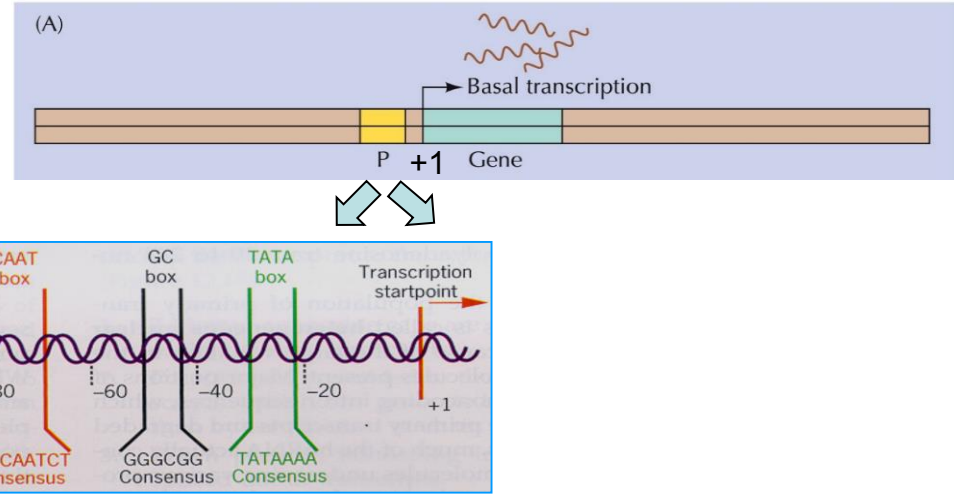
2. Transcription and its regulation in eukaryotes

2.2. RNA polymerase II and transcription factors (TFs) in eukaryotes

► Regulatory (DNA) elements for transcription initiation

- **Promoter (P) and its role**

  - locates upstream of the transcription start site (nucleotide) (+1) of a gene being transcribed.
  - influences basal or stimulatory activity of the transcription of a gene
  - there are various basal or stimulatory promoter regions, depending on the types of genes.



- **Enhancer (E) and its role**

  - can locate upstream as well as downstream of (+1)
  - Massively increases (enhances) the transcription of a gene that has enhancer

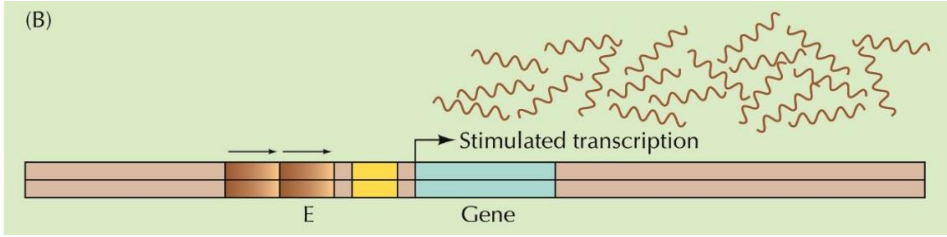
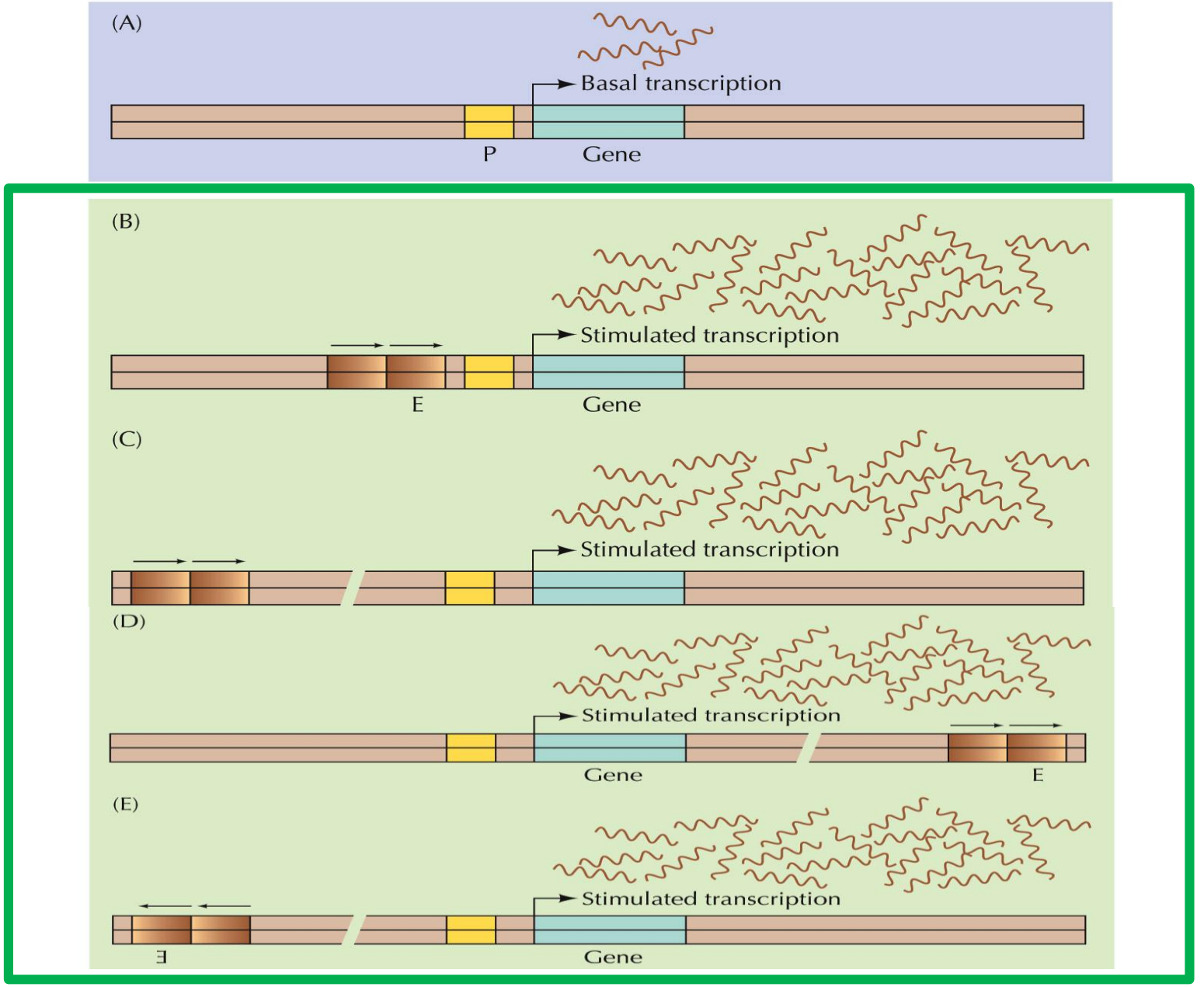


Figure 8.17 Position and action of Enhancers



- **General (or basal) promoters of an eukaryotic gene**



-BRE, TFIIB recognition elements
-Inr, the initiator elements
-DCE, MTE, DPE

- **Specific *cis*-acting regulatory DNA sequences (elements)** in controlling expression of eukaryotic genes: **Specific promoter(s), enhancer(s)**

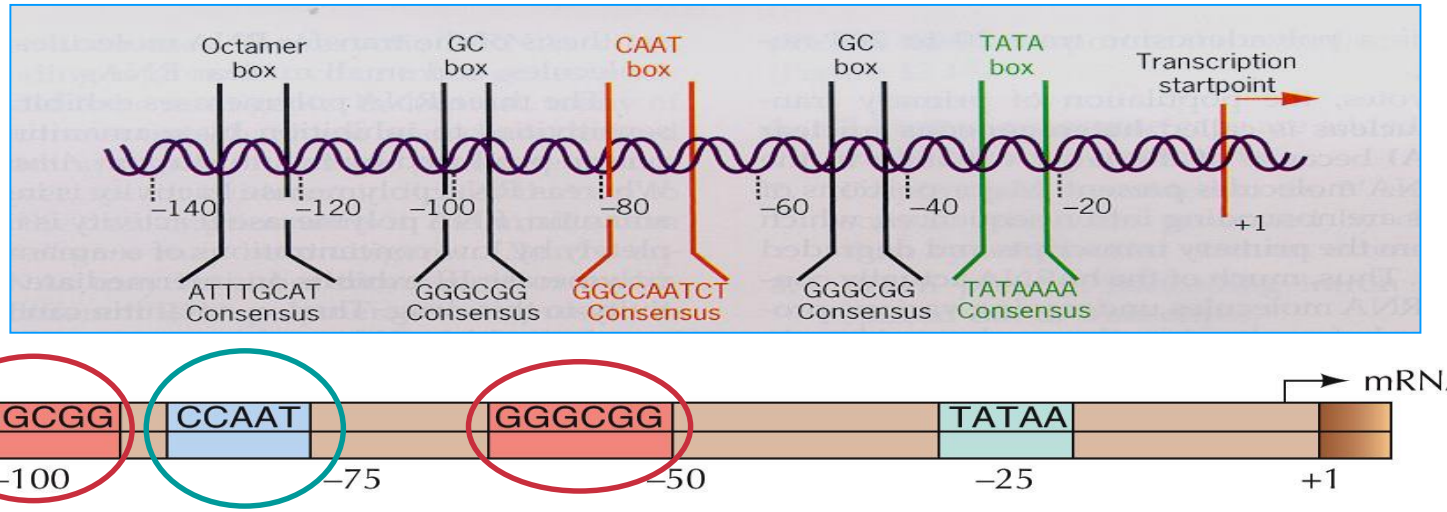


Figure 8.15 A eukaryotic promoter (eg, the herpes simplex virus gene encoding thymidine kinase)

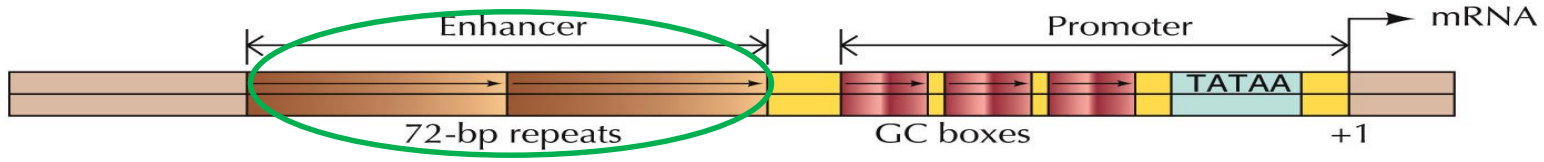
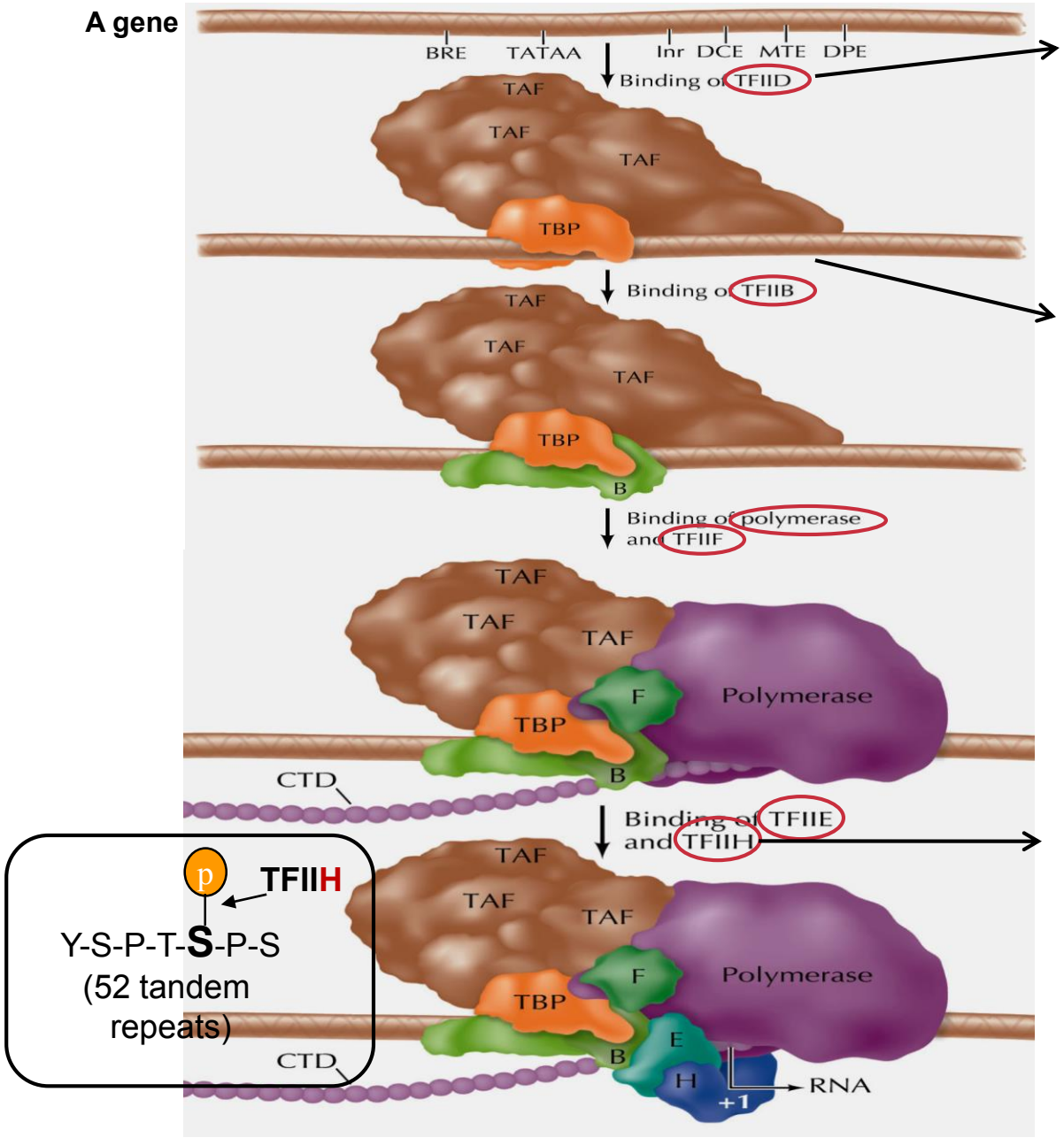


Figure 8.16 The SV40 virus gene enhancer

Figure 8.10 Formation of a polymerase II preinitiation complex *in vitro*



•TFIID

-is composed of multiple subunits, including TBP and TAFs

-TBP binds to specifically the TATA box

-TAFs bind to the Inr, DPE, DPC, and MTE

(TAF, TBP-associated factor, 14 TAFs)

•TFIIB

-binds TBP and the BRE

-also serves as a bridge to RNA pol II, which binds to the TBP-TFIIB complex in association with a 3rd general TF, **TFIIF**

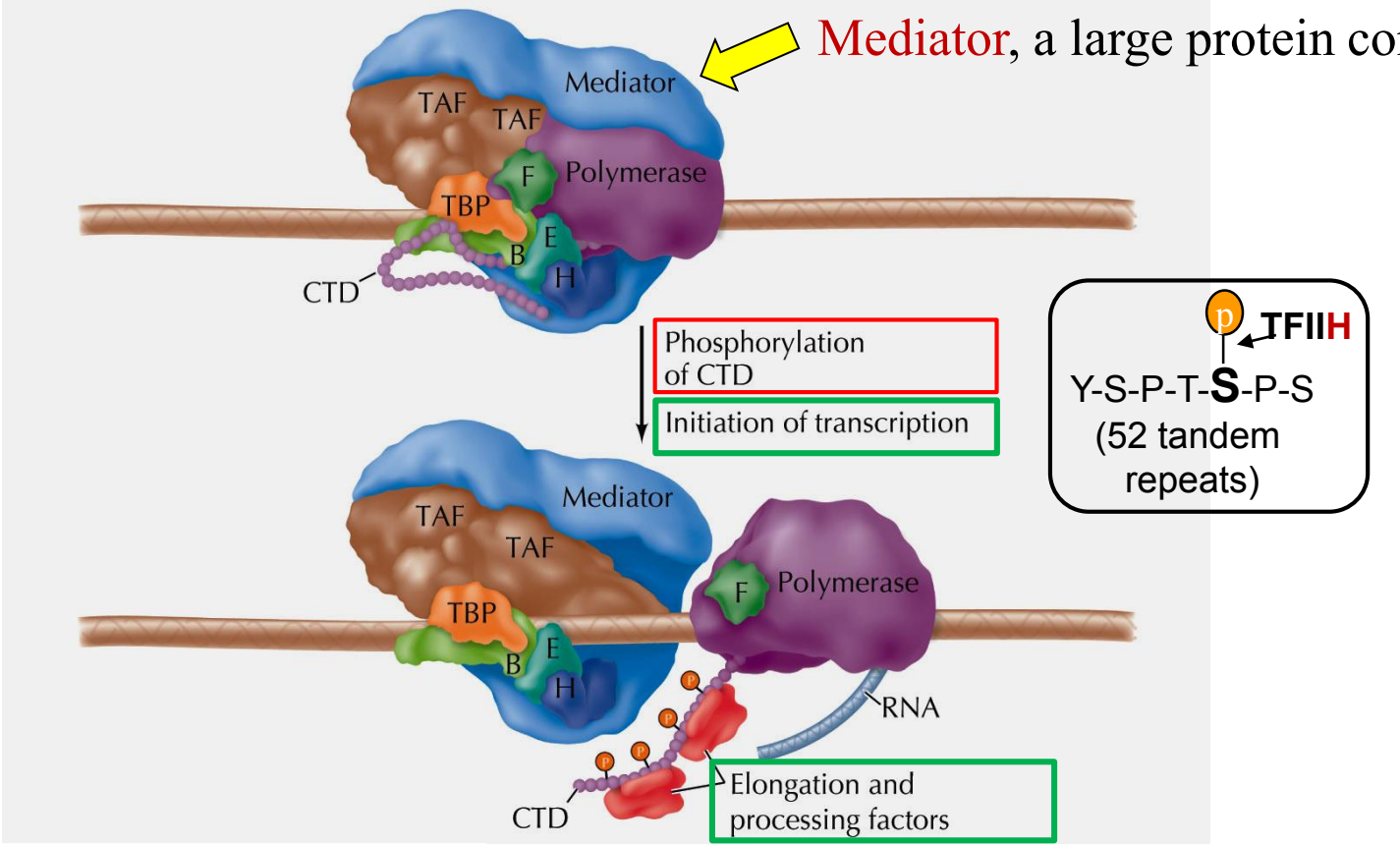
•TFIIH

-a multisubunit factors having two key roles

-has the helicase activity that unwinds DNA

-has a kinase activity to phosphorylate RNA pol II, which enables it to release from the preinitiation complex leading to the initiation of transcription

Figure 8.11 RNA Polymerase II / mediator complexes and transcription



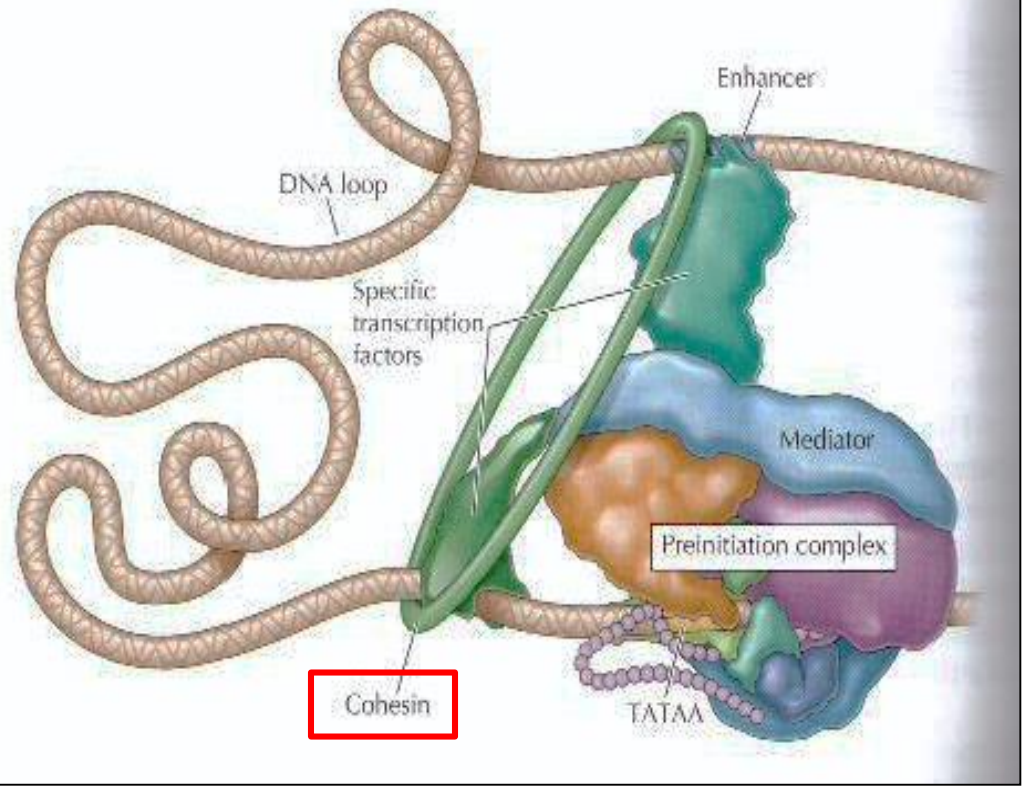
<Figure 8.11 RNA pol II / Mediator complexes and transcription

-Within a cell, **additional factors including mediator protein** are required to initiate transcription. **Mediator protein** is a large protein complex of more than 20 distinct subunits that interact with both general TFs and RNA pol II (its CTD).

-**Enhancers**, like promoters, function by binding transcription factors (TFs) that regulate RNA polymerase.

-**How enhancers work? DNA looping** allows a TF bound to a distant enhancer to interact with proteins associated with the RNA polymerase/Mediator complex at the promoter.

Figure.18 DNA looping
Transcription factors bound at distant enhancers are able to interact with Mediator or general transcription factors at the promoter because the intervening DNA can form loops, stabilized by cohesin. There is therefore no fundamental difference between the action of transcription factors bound to DNA just upstream of the gene in the promoter and to distant enhancers.



→ Regulation of transcriptional elongation

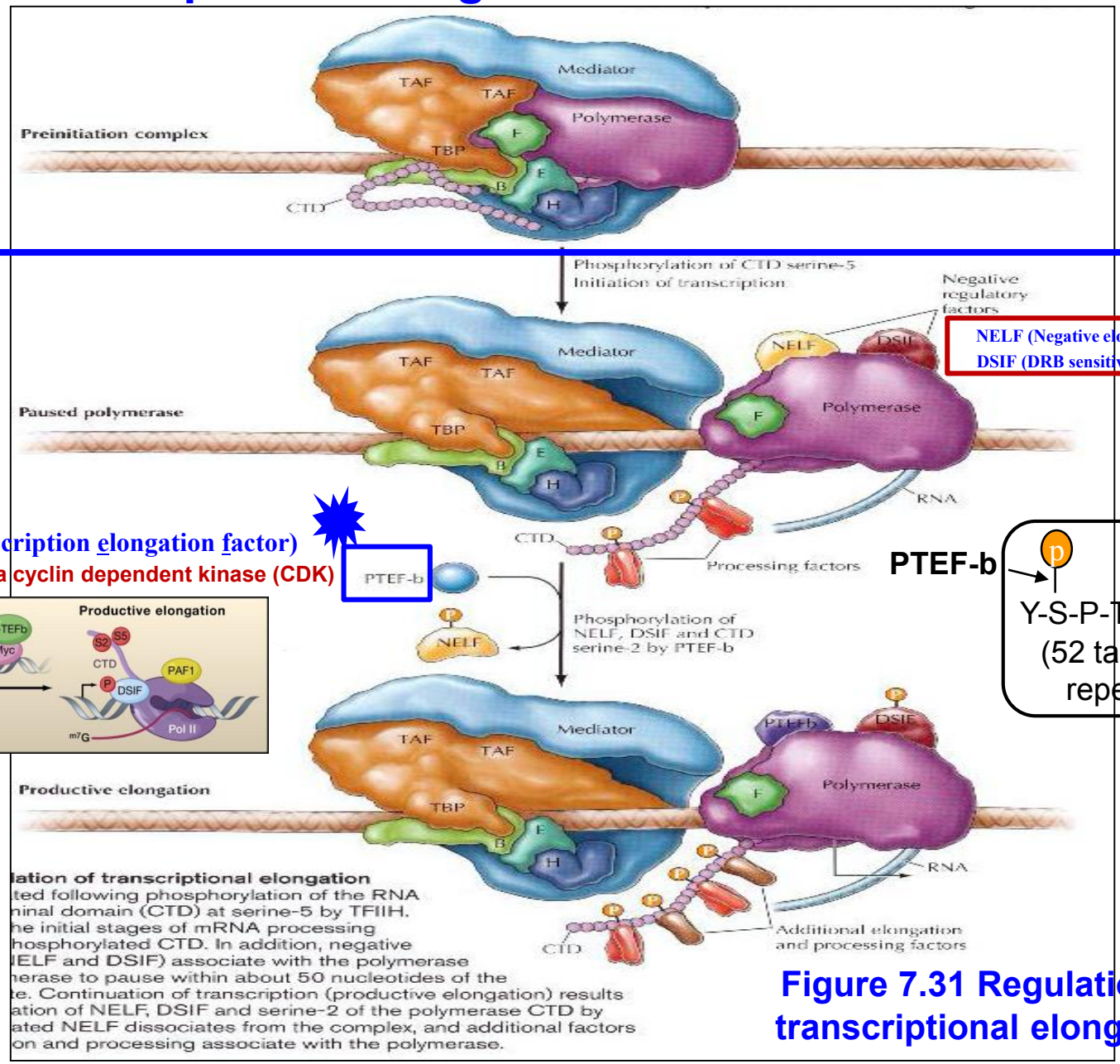


Figure 7.31 Regulation of transcriptional elongation

→ Chromatin and Epigenetics

■ Relationship of chromatin structure to transcription

-The packaging of eukaryotic DNA in chromatin has important consequences for transcription.
-As a result, the structure of chromatin is a critical aspect of gene expression.
-Indeed, actively transcribed genes are found in relatively decondensed chromatin.

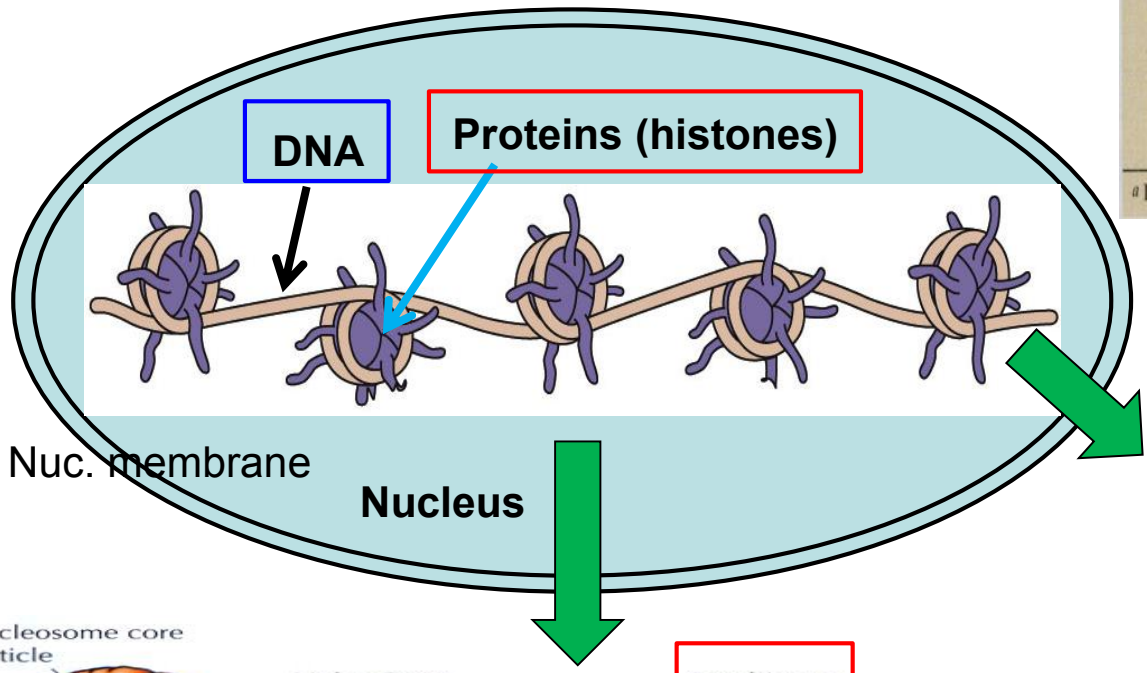


Figure 8.30 **Decondensed chromosome regions in *Drosophila.***

→ Chromatin and Epigenetics

■ Chromatin structure

-DNA + proteins (histone proteins, nonhistone proteins)



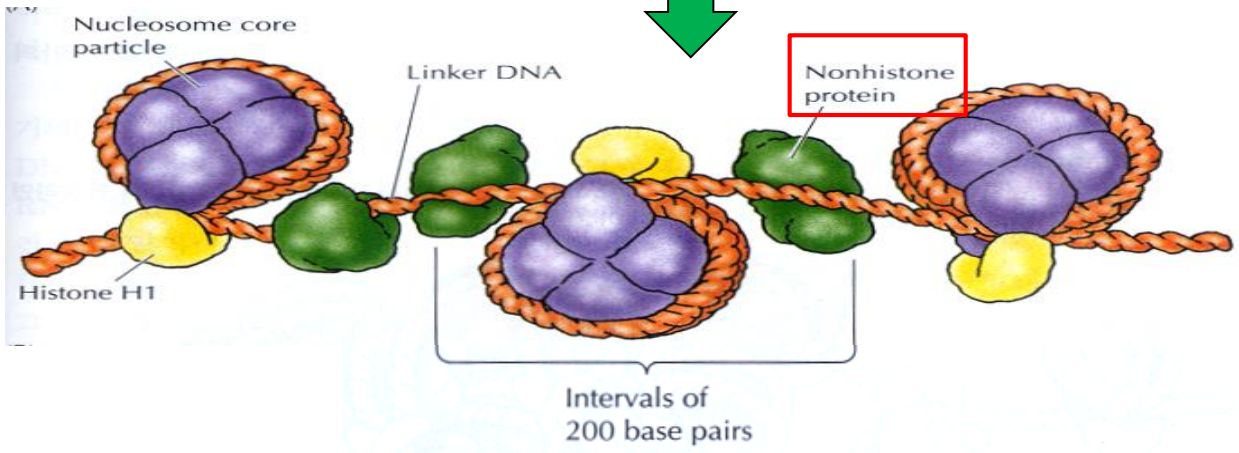
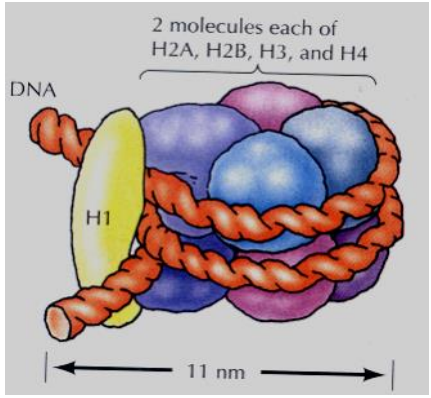


TABLE 4.3 The Major Histone Proteins			
Histone[a]	Molecular weight	Number of amino acids	Percentage Lysine + Arginine
H1, H5	22,500	244	30.8
H2A	13,960	129	20.2
H2B	13,774	125	22.4
H3	15,273	135	22.9
H4	11,236	102	24.5

[a] Data are for rabbit (H1) and bovine histones.

*** Linker histone : H1, H5**

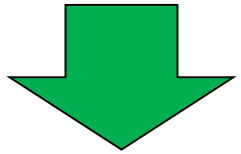
*** Core histones : H2A, H2B, H3, H4**



"The nucleosome"
(147 bp DNA)

→ Chromatin and Epigenetics

-The tight winding of DNA around the nucleosome core particle that affects both the ability of TFs to bind DNA (promoters) and the ability of RNA pol to transcribe through a chromatin template is a major obstacle to transcription.



-Chromatin structure should be modified in proper transcription.

- It can be modified by:

1) High mobility group (HMG) proteins
2) PTMs of histones (Histone protein modifications)
3) Rearrangements of nucleosomes by chromatin remodeling factors (CRFs)

→ Chromatin and Epigenetics

1) HMG proteins

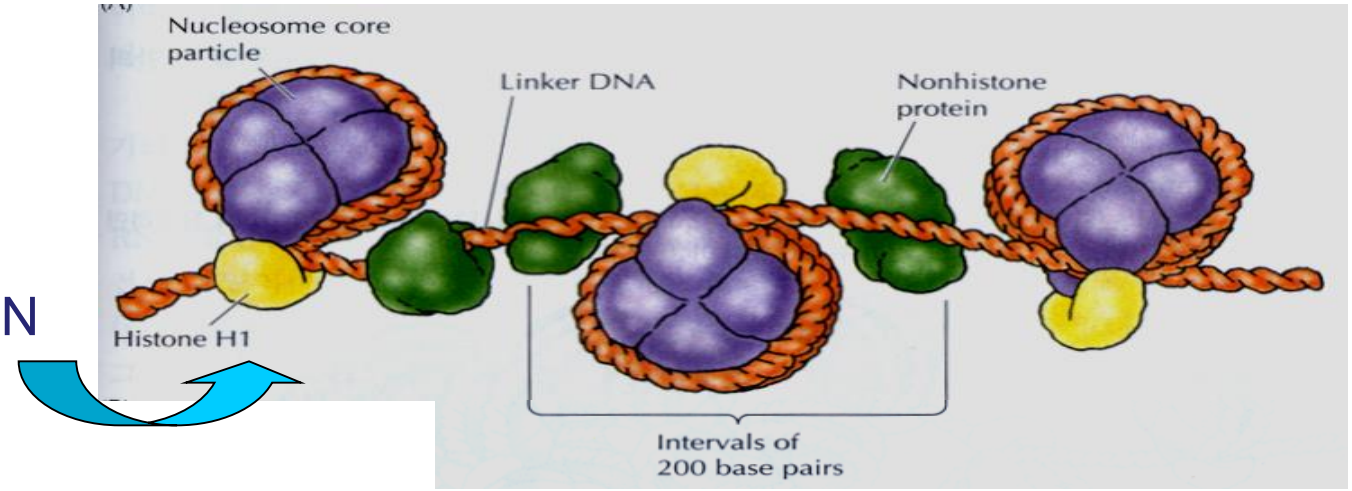
a) A superfamily of proteins that affect the architecture of chromatin structure

b) There are three family of proteins: **HMG A, B, N**

-**HMG A** and **B** **bend DNA** and thereby facilitate binding of regulatory factors to chromatin

-**HMG N** binds specifically to nucleosomes at sites that overlap the binding site of histone H1 and **induces chromatin unfolding** and **stimulates transcription by maintaining a decondensed chromatin structure**.

HMG N



→ Chromatin and Epigenetics

2) Histone proteins and PTMs

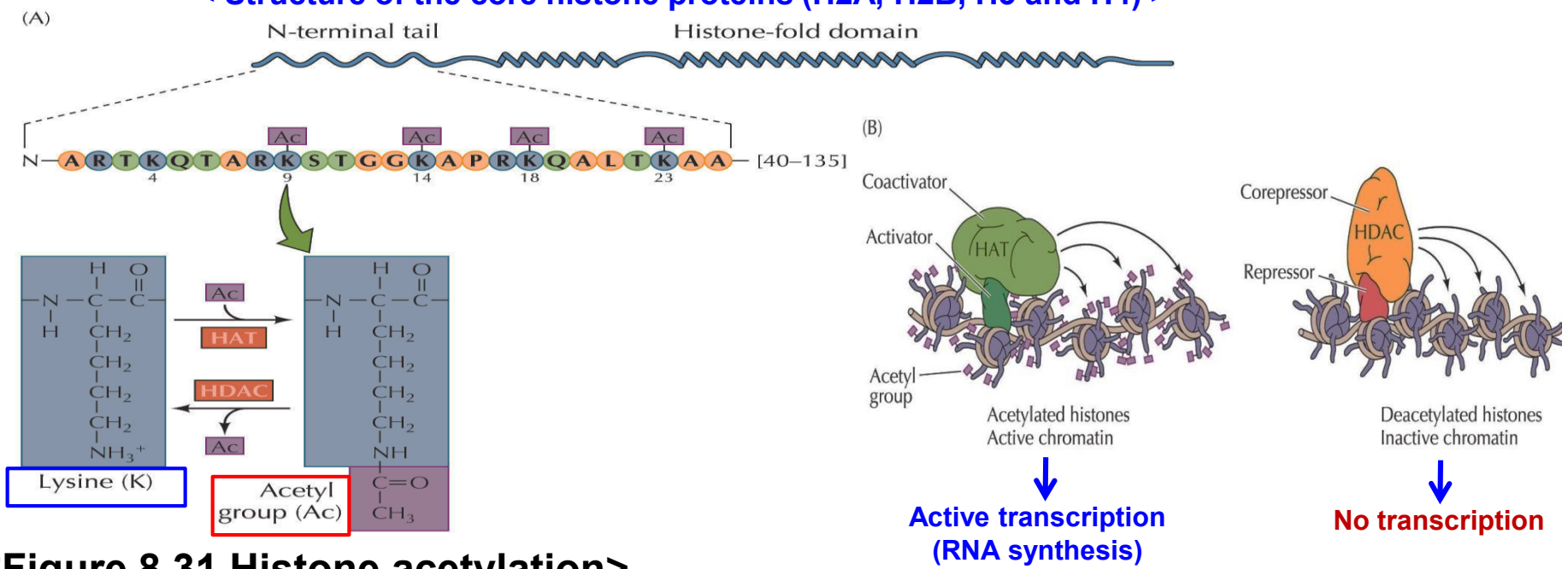
-Are the chief protein components of chromatin : They facilitate packaging the large genomes (DNA) inside the cell nucleus.

-Have many post-translational modifications (PTMs) on their N- or C-terminal tails:

1) Methylation, 2) Acetylation, 3) Phosphorylation, 4) Ubiquitination, etc.

-Such PTMs on histone proteins alter their interaction with DNA and nuclear proteins (eg, TFs)

< Structure of the core histone proteins (H2A, H2B, H3 and H4) >

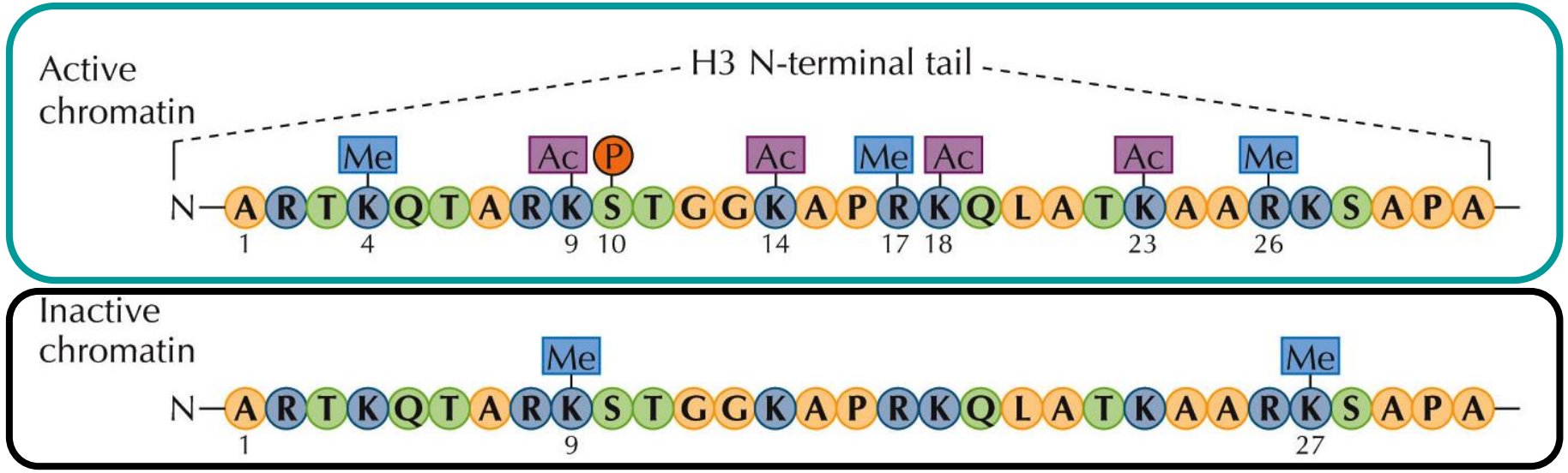


<Figure 8.31 Histone acetylation>

→ Chromatin and Epigenetics

• Types of histone PTMs and their role in transcriptional regulation

<Figure 8.32 Patterns of histone modifications>

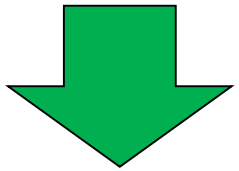


- Methylation of K4, R17, and R26 is associated with transcriptional activation.
- In contrast, methylation of K9 and K27 is linked to transcriptional repression and chromatin condensation.
- Phosphorylation of S10 facilitates transcriptional activation.
- Addition of small peptides (ubiquitin or SUMO) to K residues.

Cf. Amino acid (residue): lysine (K), arginine (R), serine (S)

→ Chromatin and Epigenetics

-The tight winding of DNA around the nucleosome core particle that affects both the ability of TFs to bind DNA (promoters) and the ability of RNA pol to transcribe through a chromatin template is a major obstacle to transcription.



-**Chromatin structure should be modified in proper transcription**.

- It can be modified by:

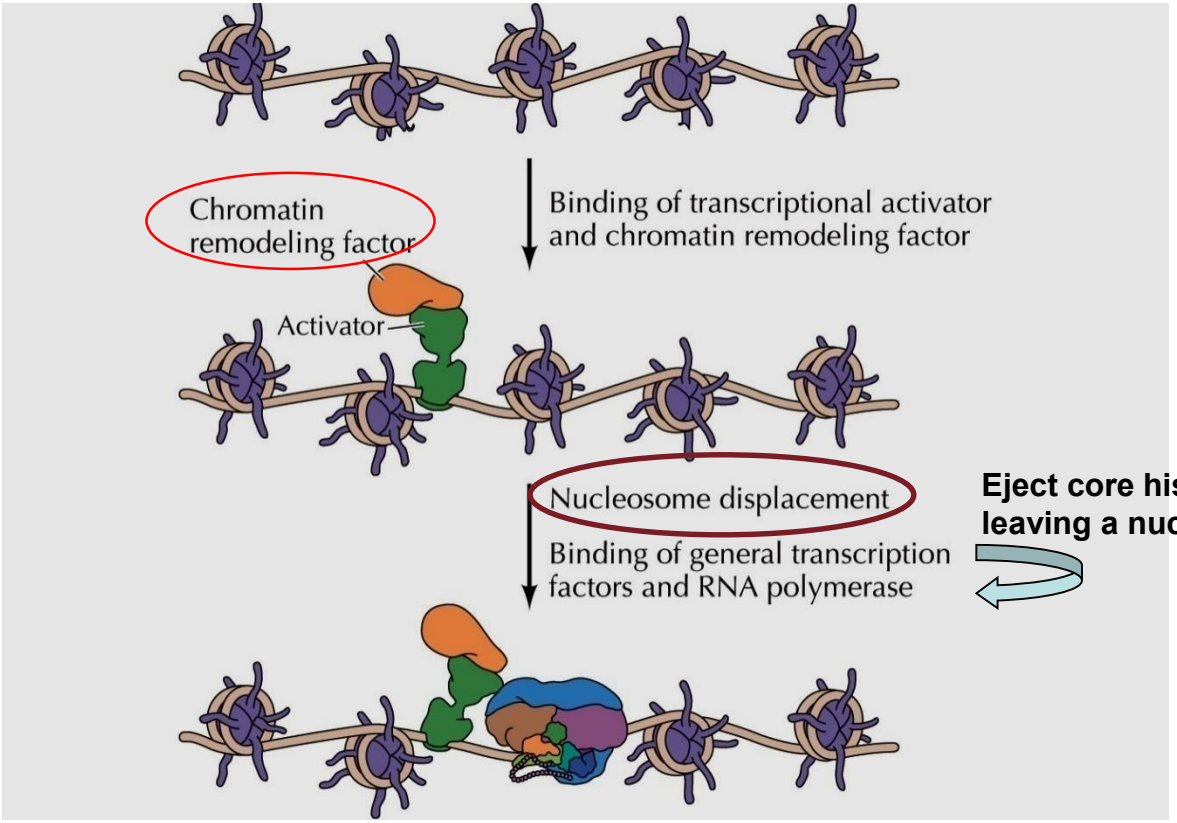
1) High mobility group (HMG) proteins
2) PTMs of histones (Histone protein modifications)
3) **Rearrangements of nucleosomes by chromatin remodeling factors (CRFs)**

→ Chromatin and Epigenetics

3) Chromatin remodeling factors (CRFs)

-Are **protein complexes that alter contacts between DNA and histones**.

-**can reposition nucleosomes**, change the conformation of nucleosomes, or eject nucleosomes from the DNA.



<Figure 8.34 Chromatin remodeling factors>

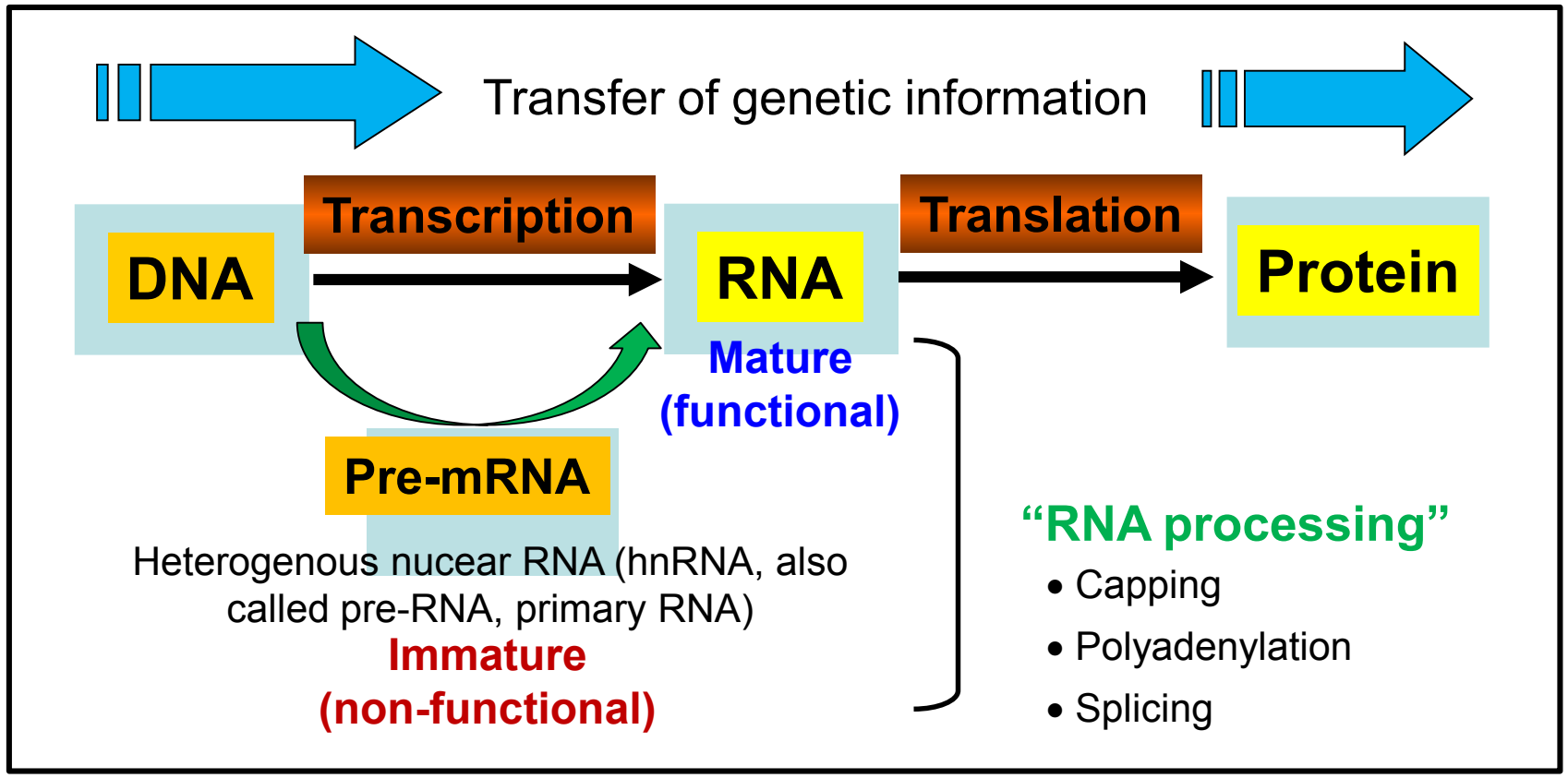
Chapter 8. RNA Synthesis and Processing 2,3

Lecture Contents

1. Introduction about learning performance of today's lecture
2. Transcription and its regulation in eukaryotes
3. Major RNA processing in eukaryotes
4. RNA turnover in eukaryotes

RNA Processing and Turnover in Eukaryotes

-Most newly-synthesized RNAs in eukaryotes must be processed or modified! (cf. bacterial RNAs are used immediately for protein synthesis while still being transcribed).



→ Regulation of RNA processing steps also provides another level of control of gene expression.

• Major RNA processing (eg, mRNA)

-In eukaryotes, **pre-mRNAs** are extensively modified before export from the nucleus.

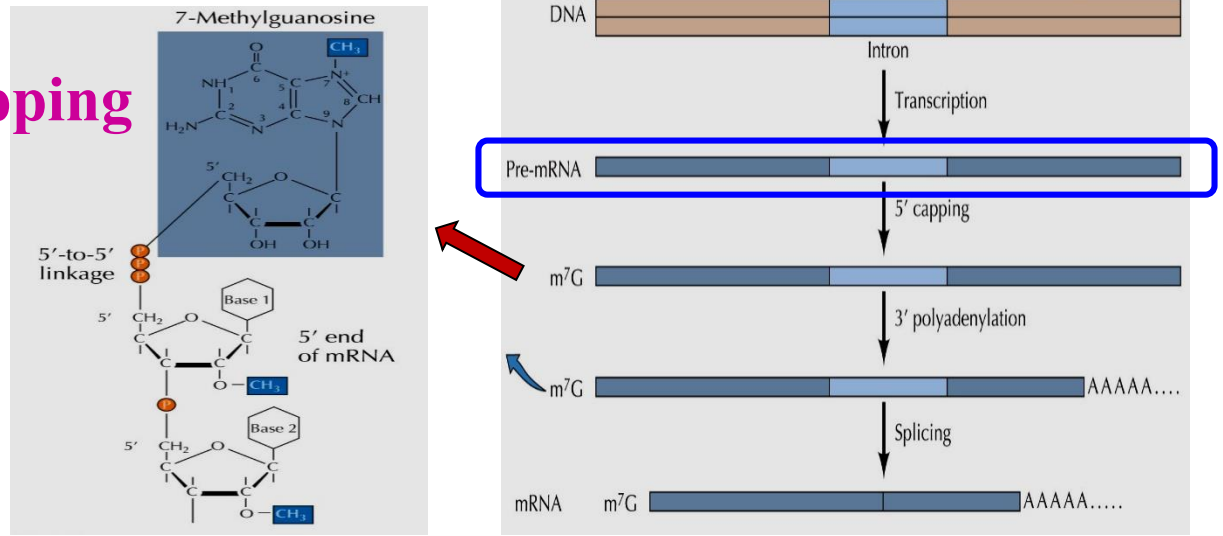
-Transcription (mRNA synthesis) and mRNA processing are coupled (see the right figure below).

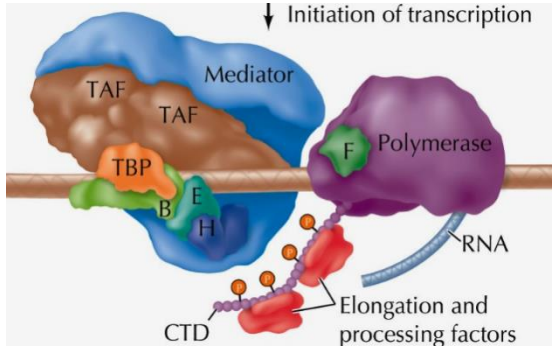
1) **Capping** - **The 5′ end** of mRNA (transcript) is modified by addition of **a 7-methylguanosine cap** (The enzymes responsible for capping are recruited to the phosphorylated CTD following transcription initiation, and the cap is added after transcription of the first 20 to 30 nucleotides).

2) **Polyadenylation** -**The 3′ end** of the transcript is modified by **addition of poly(A)**

3) **Splicing** - Noncoding RNA region, called **intron**, in the transcript is **removed** by splicing

5’ Capping





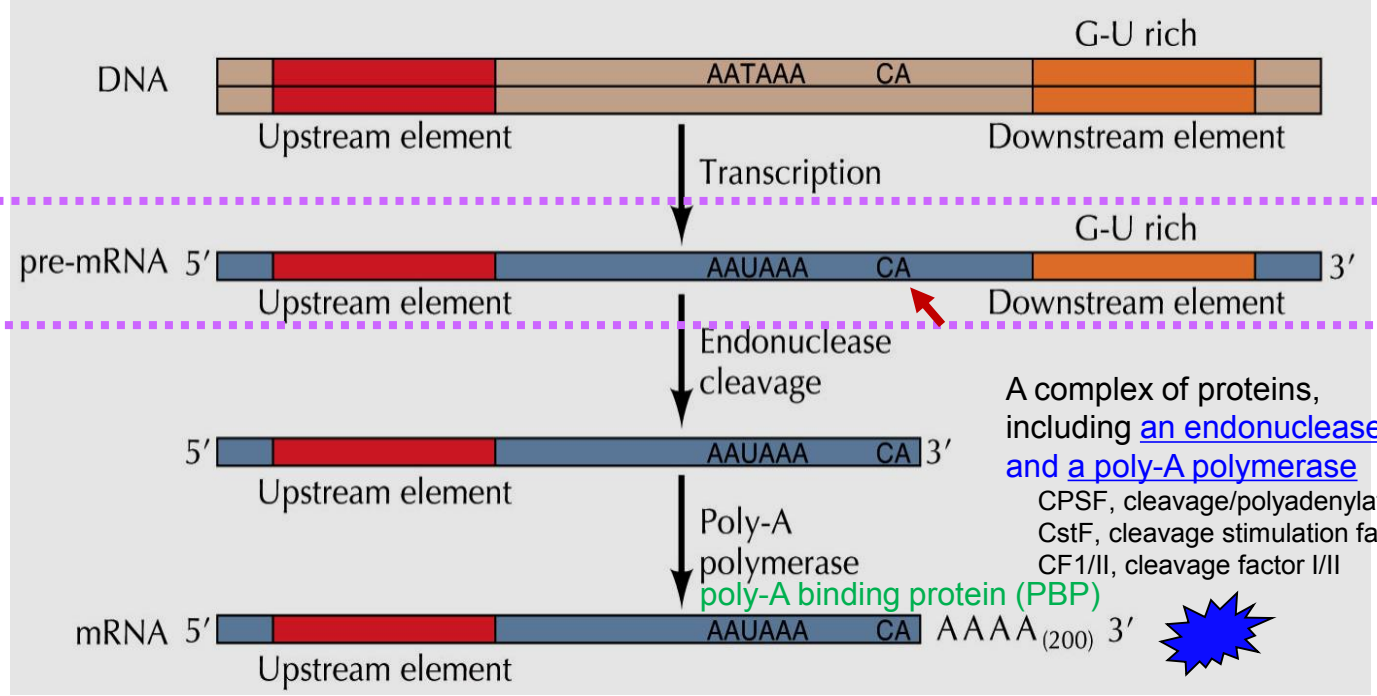
<Figure 8.44> Processing of eukaryotic messenger RNAs

• 3′ polyadenylation

-At the 3′ end, a poly-A tail is added.

-**The signals for 3′ polyadenylation** include a highly conserved **hexanucleotide (AAUAAA** in mammalian cells), and **a G-U rich** downstream sequence element.

-3′ end cleavage and polyadenylation results in termination of transcription.



<Figure 8.45> Formation of the 3′ ends of eukaryotic mRNAs

- **Splicing** :

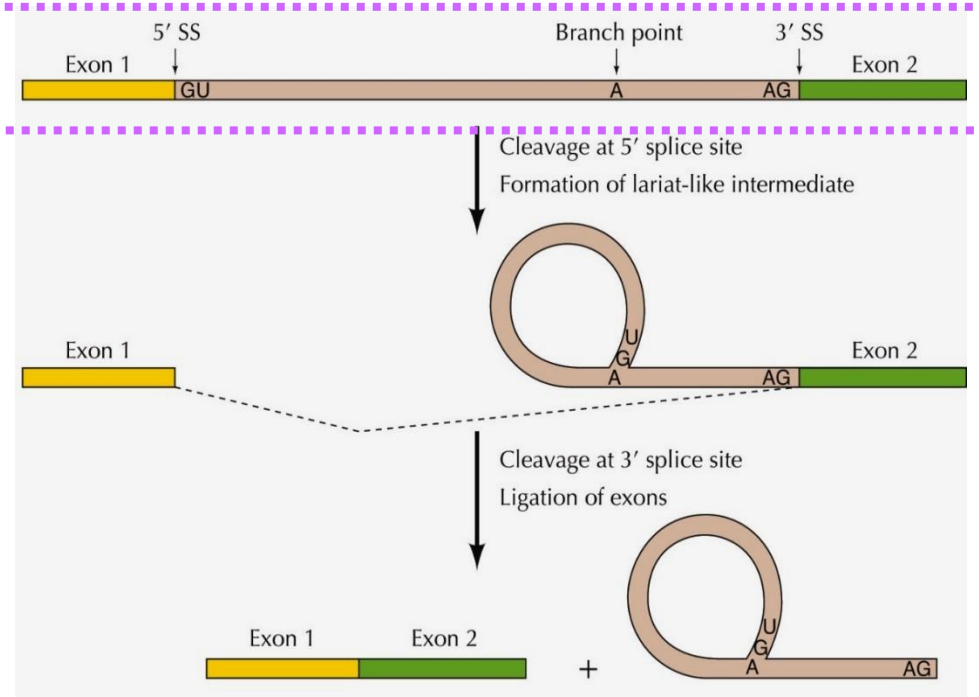
It is a process to remove introns (protein non-coding regions) from the pre-mRNA (cf. Exons : the protein coding regions from it)

- Splicing proceeds in two steps (by the action of **spliceosome**):

a) Cleavage at the 5′ splice site (SS) and joining of the 5′ end of the intron to an A within the intron (the branch point). The intron forms a loop.

b) Cleavage at the 3′ splice site and simultaneous ligation of the exons, resulting in excision of the intron loop.

Pre-mRNA



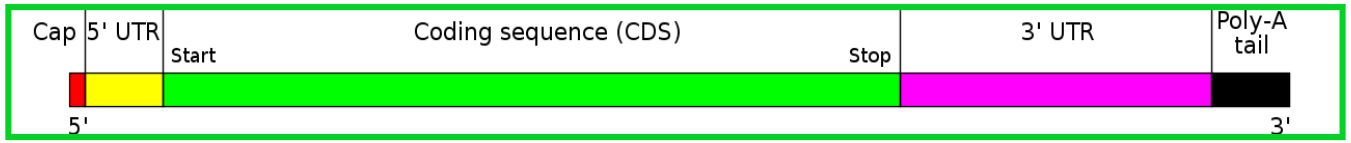
"Spliceosome"

→ A large complex with proteins and RNAs

-snRNAs: U1,U2,U3,U5, U6 (each size: 50-200 bp)

-are complexed with 6 to 10 proteins to form **small nuclear ribonucleoprotein particle (snRNP)**, which involves in the splicing process

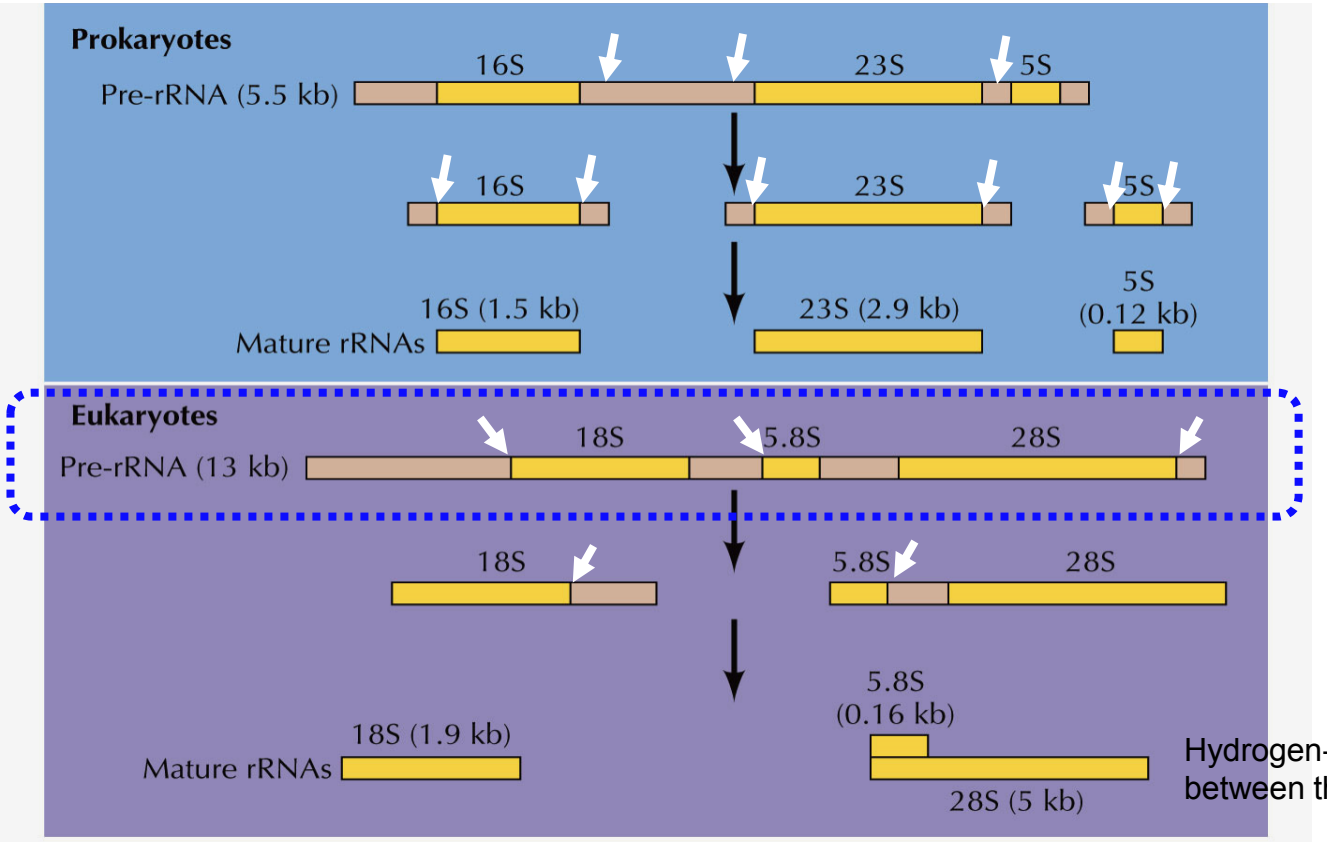
<Figure 8.47> Splicing of pre-mRNA

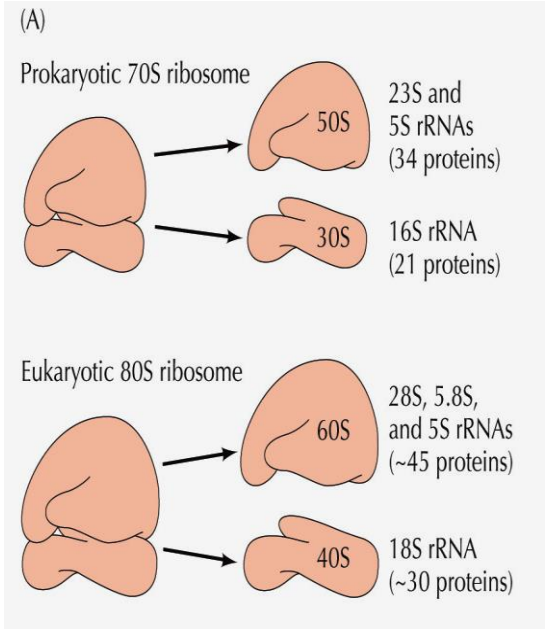


• Processing of rRNAs

-**The ribosomal RNAs of both prokaryotes and eukaryotes** are derived from **a single long pre-rRNA molecule**.

-**In eukaryotes, <u>5S rRNA</u> is transcribed from a separate gene**.

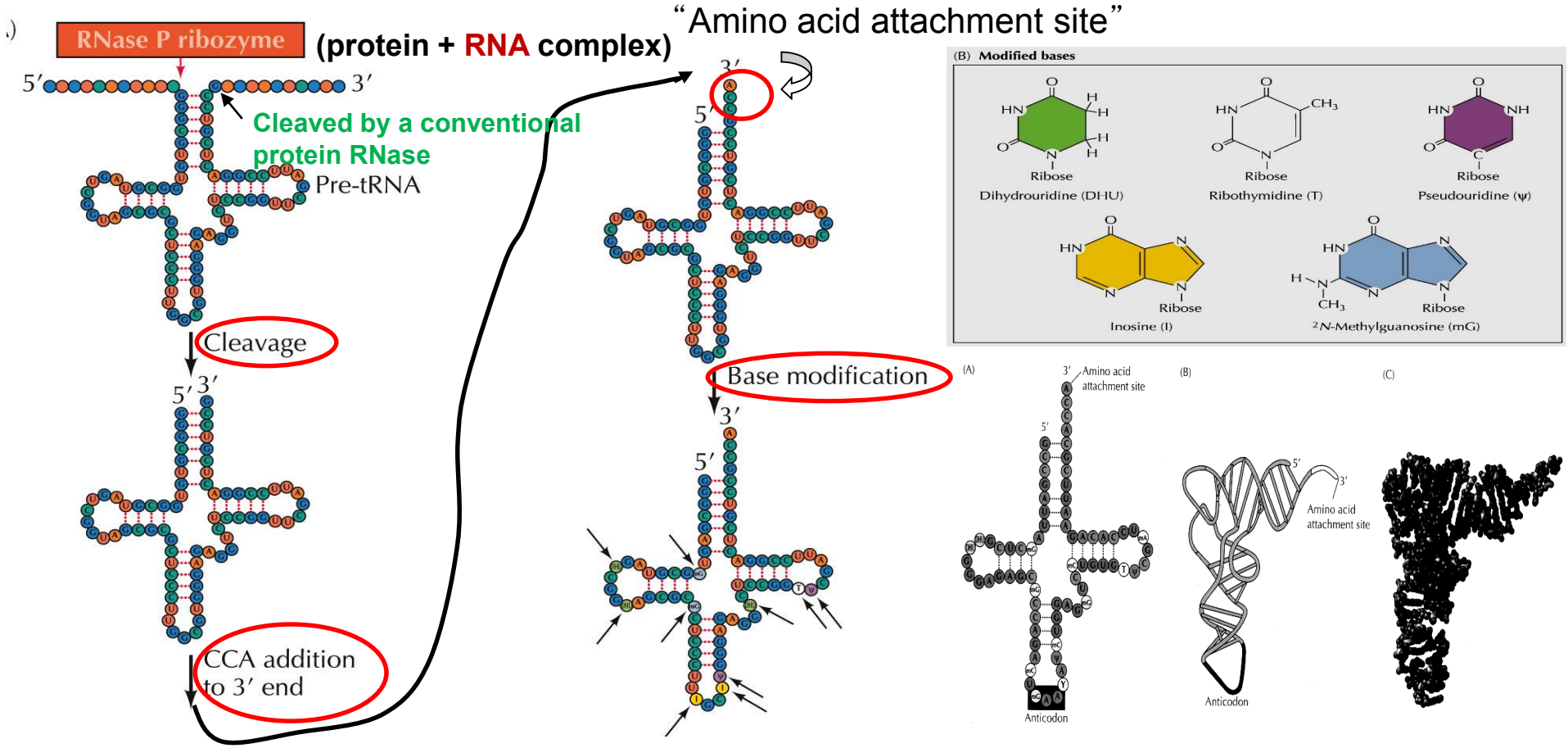




Hydrogen-bonded between these rRNAs

<Figure 8.42> Processing of ribosomal RNAs>

• Processing of tRNAs



<Figure 8.43> Processing of transfer RNAs

• RNA editing

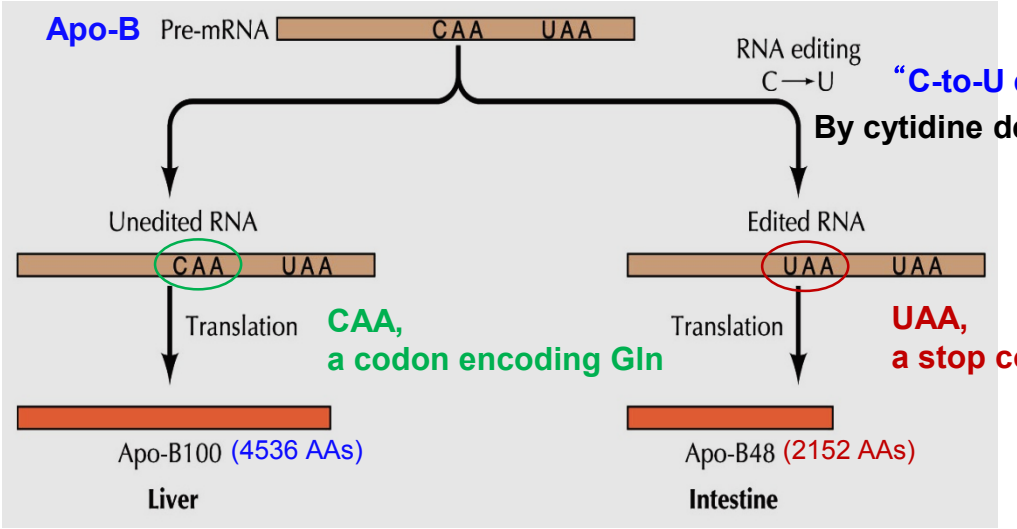
-Is **a processing that alters the protein-coding sequences of some mRNAs**.

-It involves **single base modification reactions** such as **deamination of cytosine to uridine and adenosine to inosine**.

-Editing of the mRNA for **apolipoprotein B (Apo-B)**, **which transports lipids in the blood**, results in two different proteins:

- **Apo-B100** is synthesized in **the liver** by translation of **the unedited mRNA**.
- **Apo-B48** is synthesized in **the intestine** from edited mRNA in which **a C has been changed to a U** by deamination.

<Figure 8.54> Editing of apolipoprotein B mRNA



"Tissue-specific editing" → structurally and functionally different expression of apolipoprotein Bs !!

Chapter 8. RNA Synthesis and Processing 2,3

Lecture Contents

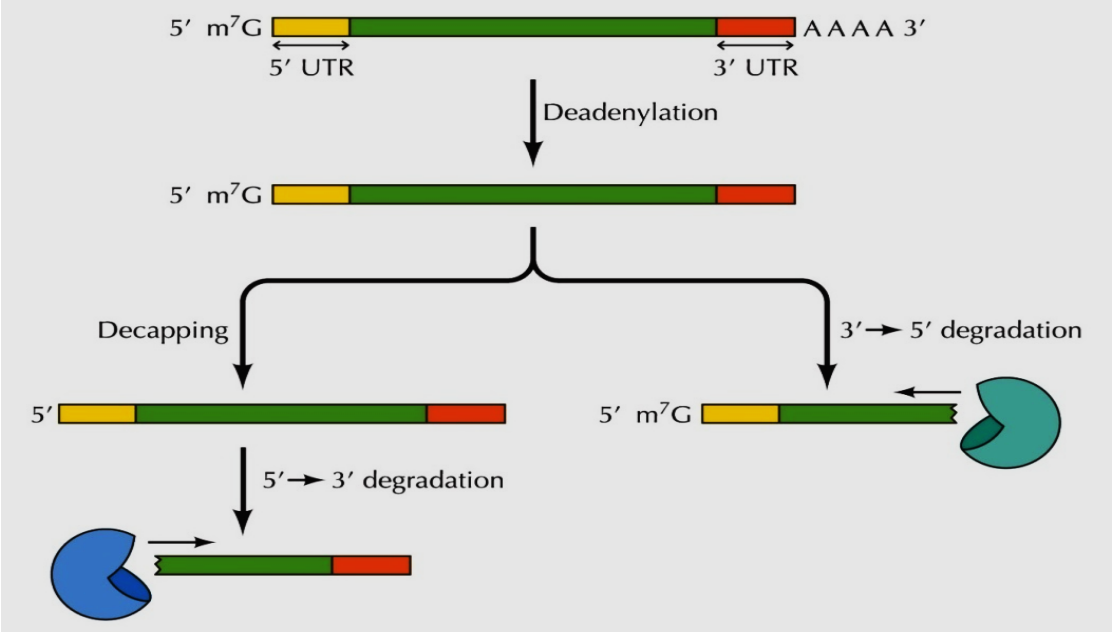
1. Introduction about learning performance of today's lecture
2. Transcription and its regulation in eukaryotes
3. Major RNA processing in eukaryotes
4. RNA turnover in eukaryotes

RNA turnover in Eukaryotes

• mRNA degradation (turnover)

-Degradation of eukaryote mRNAs is initiated by shortening of poly-A tails.

<Figure 8.55>
mRNA degradation



-Rapidly degraded mRNAs often contain specific AU-rich sequences (ARE) near the 3′ ends which are binding sites for proteins that can either stabilize them or target them for degradation.

-The RNA-binding proteins (eg, HuR or tristetraproline (TTP)) are regulated by extracellular signals, such as growth factors and hormones.

-Degradation of some mRNAs is regulated by both siRNAs and miRNAs.

Chapter 8. RNA Synthesis and Processing 2,3

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1. Introduction about learning performance of today's lecture
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다음 시간!

Chapter 9
Protein Synthesis and Processing 1

Thanks for your attention!